

Understanding Statistical Learning Theory (MA6056)

PREREQUISITES

July – November 2026

Sampling Distributions based on Normal Population

Univariate Normal Distribution

Definition 1: A continuous random variable X is said to have a univariate normal distribution if the PDF of X is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \text{ for all } x \in \mathbb{R},$$

where $\mu \in \mathbb{R}$ and $\sigma > 0$.

Notation: $X \sim N(\mu, \sigma^2)$.

Theorem 1: If $X \sim N(\mu, \sigma^2)$, all moments of X exist. In particular, $E(X)$ and $Var(X)$ exist, and they are given by $E(X) = \mu$ and $Var(X) = \sigma^2$.

Remark 1: This means that a normal distribution is completely specified by its mean and variance.

Bivariate Normal

Definition 2: A two dimensional random vector $\mathbf{X} = (X_1, X_2)$ is said to have a bivariate normal distribution if $aX_1 + bX_2$ is a univariate normal for all $(a, b) \in \mathbb{R}^2 \setminus (0, 0)$.

Remark 2: If $\mathbf{X}$ has bivariate normal distribution, then each of X_1 and X_2 is univariate normal. Hence $E(X_1)$, $E(X_2)$, $Var(X_1)$, $Var(X_2)$, and $Cov(X_1, X_2)$ exist.

Notation: $\boldsymbol{\mu} = E(\mathbf{X}) = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$ and $\Sigma = Var(\mathbf{X}) = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}$.

Theorem 2: Let $\mathbf{X}$ be a bivariate normal random vector. If $\boldsymbol{\mu} = E(\mathbf{X})$ and $\Sigma = Var(\mathbf{X})$, then for any fixed $\mathbf{u} = (a, b) \in \mathbb{R}^2 \setminus (0, 0)$, $\mathbf{u}'\mathbf{X} \sim N(\mathbf{u}'\boldsymbol{\mu}, \mathbf{u}'\Sigma\mathbf{u})$.

Bivariate Normal

Theorem 3: Let $\mathbf{X}$ be a bivariate normal random vector, then $M_{\mathbf{X}}(\mathbf{t}) = e^{\mathbf{t}'\boldsymbol{\mu} + \frac{1}{2}\mathbf{t}'\boldsymbol{\Sigma}\mathbf{t}}$ for all $\mathbf{t} \in \mathbb{R}^2$.

Remark 3: Thus the bivariate normal distribution is completely specified by the mean vector $\boldsymbol{\mu}$ and the variance-covariance matrix $\boldsymbol{\Sigma}$.

Notation: $\mathbf{X} \sim N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Corollary 1: If $\mathbf{X} \sim N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $X_1 \sim N(\mu_1, \sigma_{11})$ and $X_2 \sim N(\mu_2, \sigma_{22})$.

Remark 4: The converse of the above theorem is not true.

Remark 5: If $\mathbf{X} \sim N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $Cov(X_1, X_2) = 0$, then X_1 and X_2 are independent.

Probability Density Function

Theorem 4: Let $\mathbf{X} \sim N_2(\boldsymbol{\mu}, \Sigma)$ be such that Σ is invertible, then, for all $\mathbf{x} \in \mathbb{R}^2$, $\mathbf{X}$ has a joint PDF given by

$$\begin{aligned} f(\mathbf{x}) &= \frac{1}{2\pi|\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})' \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\} \\ &= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{A(x_1, x_2, \mu_1, \mu_2, \sigma_1, \sigma_2, \rho)}, \end{aligned}$$

where $\sigma_1 = \sqrt{\sigma_{11}}$, $\sigma_2 = \sqrt{\sigma_{22}}$, ρ is correlation coefficient between X_1 and X_2 , and

$$A = -\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{x_1 - \mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x_1 - \mu_1}{\sigma_1} \right) \left(\frac{x_2 - \mu_2}{\sigma_2} \right) + \left(\frac{x_2 - \mu_2}{\sigma_2} \right)^2 \right\}.$$

Conditional Probability Density Function

Theorem 5: Let $\mathbf{X} \sim N_2(\boldsymbol{\mu}, \Sigma)$ be such that Σ is invertible, then

1. for all $x_2 \in \mathbb{R}$, the conditional PDF of X_1 given $X_2 = x_2$ is given by

$$f_{X_1|X_2}(x_1|x_2) = \frac{1}{\sigma_{1|2}\sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x_1 - \mu_{1|2}}{\sigma_{1|2}} \right)^2 \right] \quad \text{for } x_1 \in \mathbb{R},$$

where $\mu_{1|2} = \mu_1 + \rho \frac{\sigma_1}{\sigma_2}(x_2 - \mu_2)$ and $\sigma_{1|2}^2 = \sigma_1^2(1 - \rho^2)$.

2. $X_1|X_2 = x_2 \sim N(\mu_{1|2}, \sigma_{1|2}^2)$.
3. $E(X_1|X_2 = x_2) = \mu_{1|2} = \mu_1 + \rho \frac{\sigma_1}{\sigma_2}(x_2 - \mu_2)$ for all $x_2 \in \mathbb{R}$.
4. $Var(X_1|X_2 = x_2) = \sigma_{1|2}^2 = \sigma_1^2(1 - \rho^2)$ for all $x_2 \in \mathbb{R}$. Hence the conditional variance does not depend on x_2 .

Dist. of Sample Mean and Variance

Theorem 6: Let $X_1, X_2, \dots, X_n$ be i.i.d. $N(0, 1)$ random variables. Then $\sum_{i=1}^n X_i^2 \sim \text{Gamma}(n/2, 1/2) \equiv \chi_n^2$.

Theorem 7: Let $X_1, X_2, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ random variables. Then $\bar{X} \sim N(\mu, \sigma^2/n)$, $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$, and $\bar{X}$ and S^2 are independently distributed. Here $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.

Student's t Distribution

Theorem 8: Let X and Y are two independent random variables with $X \sim N(0, 1)$ and $Y \sim \chi_\nu^2$. Then, for all $t \in \mathbb{R}$, the PDF of the random variable $T = \frac{X}{\sqrt{Y/\nu}}$ is

$$f_T(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}.$$

Remark 6: A random variable T is said to have Student's t distribution with degrees of freedom ν if the PDF of the random variable is $f_T(\cdot)$ as given above. We will use the notation $T \sim t_\nu$.

Corollary 2: Let $X_1, X_2, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ random variables. Then

$$\sqrt{n} \frac{\bar{X} - \mu}{S} \sim t_{n-1},$$

where S^2 is the sample variance.

Theorem 9: Let X and Y be two independent RVs with $X \sim \chi_{d_1}^2$ and $Y \sim \chi_{d_2}^2$. Then, for all $x > 0$, the PDF of the RV $F = \frac{X/d_1}{Y/d_2}$ is

$$f_F(x) = \frac{1}{B\left(\frac{d_1}{2}, \frac{d_2}{2}\right)} \left(\frac{d_1}{d_2}\right)^{\frac{d_1}{2}} x^{\frac{d_1}{2}-1} \left(1 + \frac{d_1}{d_2}x\right)^{-\frac{d_1+d_2}{2}}.$$

Remark 7: A random variable F is said to have F distribution with degrees of freedoms d_1 and d_2 if the PDF of the random variable is $f_F(\cdot)$ as given above. We will use the notation $F \sim F_{d_1, d_2}$ to mean that the RV F has a F distribution with d_1 and d_2 degrees of freedoms.

Corollary 3: Let $X_1, X_2, \dots, X_n$ be i.i.d. $N(\mu_1, \sigma_1^2)$ RVs. Let $Y_1, Y_2, \dots, Y_m$ be i.i.d. $N(\mu_2, \sigma_2^2)$ RVs. Also, assume that X_i 's and Y_j 's are independent RVs. Then

$$\frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \sim F_{n-1, m-1},$$

where S_1^2 and S_2^2 are sample variances based on X_i 's and Y_j 's respectively.

Point Estimation

- All most all field of study uses it.
- News papers
- Sports
- Science and Technology
- Social Sciences . . .

Aim and Issues

- Our aim is to find information regarding some numerical characteristic of a collection of items/persons/product. This collection is called **population**.
- Most of the time it is not possible to enumerate each and every individual in the population. For example, it is very costly (in terms of money and time) to go to every Indian and note down their height to calculate the average.
- In many other cases, it is not possible to enumerate each and every item in the population. For example, suppose that we want to find the average lifetime of an electronic item manufactured by a company. Here we cannot put all the items manufactured by the company on test and wait for failure of all the items.

Approach

- Take a subset of the population based on which we try to find out the value of the numerical characteristic. This subset is called a **sample**. Obviously, the obtained value will not be exact, hence it is an estimate.
- We should choose the sample such that it will be a good representative of the population. Otherwise the estimate may not be close (in some sense) to the original value, which we do not know.
- There are different ways of selecting sample from a population. We will not discuss this issue here.
- We will consider one such sample which is called random sample (definition will be given).

Statistical Modeling

- As the value of numerical characteristic may vary from item to item of the population, we will model it with a random variable and the uncertainty using a probability distribution.
- Let X be a random variable (either DRV or CRV) which denotes the numerical characteristic of the population.
- There are two possibilities:
 1. X has a CDF F with known functional form except perhaps some parameters. Here our aim is to find the (educated) guess value of the parameters. This is known as **parametric inference**.
 2. X has a CDF F whose functional form is unknown. This is known as **nonparametric inference**.
- In this course we will mainly consider parametric inference.

Parametric Inference

- We will assume that the class of distribution is indexed by parameter θ . It could be scalar or vector valued. The parameter θ is fixed but unknown. We also assume that the possible values of θ belongs to a set Θ . Θ is called **parametric space**.
- **Example 1:** $X \sim N(\mu, \sigma^2)$. Here $\mu \in \mathbb{R}$ and $\sigma > 0$. We can take $\theta = (\mu, \sigma)$ and $\Theta = \mathbb{R} \times \mathbb{R}^+$.
- **Example 2:** Let us assume that X denotes the height of a large community of people and we try to model it using a normal distribution. It is plausible to assume that $\mu > 0$. Hence Θ would be $\mathbb{R}^+ \times \mathbb{R}^+$.
- **Aim:** To decide which member of the family $\{F_\theta : \theta \in \Theta\}$ can represent the CDF of X , which is equivalent to decide the value of θ in Θ .

Definition 3: The RVs $X_1, X_2, \dots, X_n$ is said to be a random sample (RS) of size n from the population F if $X_1, X_2, \dots, X_n$ are independently and identically distributed with marginal CDF F .

Remark 8:

- F may (and for inference will) depend on unknown parameters θ .
- How to model a data using RS?
 - X_i denotes the i th observation.
 - We need to make sure that the observation are taken such a way that the value of one does not effect the others.
 - And hence each X_i is an observation on the same variable and each X_i has same CDF F .

- Joint CDF of a RS $X_1, \dots, X_n$ is

$$F_{\mathbf{X}}(x_1, \dots, x_n) = \prod_{i=1}^n F(x_i).$$

- Joint PMF/PDF of a RS $X_1, \dots, X_n$ is

$$f_{\mathbf{X}}(x_1, \dots, x_n) = \prod_{i=1}^n f(x_i).$$

Definition 4: Let $X_1, \dots, X_n$ be a RS. Let $T(x_1, \dots, x_n)$ be a function having domain that includes the cartesian product of the supports of $X_1, X_2, \dots, X_n$. Then the RV $\mathbf{Y} = T(X_1, \dots, X_n)$ is called a statistic if it is not a function of unknown parameters.

Remark 9:

- Note that our aim is to find a guess value of unknown parameters based on a RS. Hence, we are considering a function of RS.
- If the function involve any unknown parameters, we will not be able to compute the value of the function given a realization of a RS. Hence, the function that involves unknown parameters is of no use in this respect.

Example 3: Let $X_1, \dots, X_n$ be a RS from a population with PDF $f_{\mu, \sigma}$, where μ and σ are both unknown. Then

- $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, $S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ are examples of statistic.
- However, $\frac{\bar{X} - \mu}{\sigma}$ is not a statistic.

Definition 5: In the context of estimation, a statistic is called a point estimator.

Remark 10:

- In the above definition, we do not mention about the parameter that is to be estimated.
- No mention of the range of the estimator.
- However, in practice we need to take care of the parameter to be estimated and its' range. For example, to estimate population variance, we should not use an estimator that can be negative.
- Estimator is a function of RS, and estimate is the realized value of an estimator.

Finding Point Estimator

- There are several methods to find an estimator.
- We will consider three of them:
 - Method of moment estimator
 - Maximum likelihood estimator
 - Least square estimator

Method of Moment Estimator (MME)

- Introduced by Karl Pearson in the year 1902.
- The method is as follows:
 1. Suppose that we have a RS of size n form a population with PMF/PDF $f(x; \theta)$, where $\theta = (\theta_1, \dots, \theta_k)$ is the unknown parameter.
 2. Calculate first k (no. of unknown parameters) moments $\mu'_1, \dots, \mu'_k$ of $f(x; \theta)$.
 3. Calculate first k sample moments $m'_1, \dots, m'_k$. Here m'_r is define by $m'_r = \frac{1}{n} \sum_{i=1}^n X_i^r$.
 4. Equate $\mu'_r = m'_r$ for $r = 1, 2, \dots, k$.
 5. Solve the system of k equations (if they are consistent) for θ_i 's. The solutions are the MMEs of the unknown parameters.

Examples

Example 4: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(\theta)$, $\theta \in [0, 1] = \Theta$. Here we have one parameter θ , with $k = 1$. $E(X_1) = \theta$. Hence we get the MME of θ is $\hat{\theta} = \bar{X}$.

Example 5: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+ = \Theta$. Here $k = 2$, $E(X) = \mu$, and $E(X^2) = \sigma^2 + \mu^2$. Hence we get the MMEs of μ and σ^2 are $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$, respectively.

Examples

Example 6: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$, $\sigma > 0$.

Here $k = 1$. However, as $E(X) = 0$, equating $E(X) = \bar{X}$ does not provide any solution (inconsistent). Alternatively we can find

$E(X^2) = \sigma^2$ and equate to m'_2 to obtain MME of σ^2 as

$$\widehat{\sigma^2} = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

Example 7: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\theta, \theta^2)$, $\theta > 0$.

Here $k = 1$. $E(X) = \theta$. Equating $E(X) = \bar{X}$, we get MME of θ is $\widehat{\theta} = \bar{X}$. However, this may not be a meaningful estimator as $\bar{X}$ can be negative with positive probability, while $\theta > 0$.

Remark 11: Previous two examples show that there are some degree of arbitrariness in this method.

Maximum Likelihood Estimator (MLE)

- Proposed by R. A. Fisher in 1912.
- One of the most popular method of estimation.
- Let us start with an example (next slide).

Example

Example 8: Let a box has some red balls and some black balls. It is known that number of black balls to red balls is in 1:1 or 1:2 ratio. We want find whether it is 1:1 or 1:2. We may proceed as follows:

- Randomly draw two balls from the box.
- Let X be the number of black balls out of two drawn balls.
- $X \sim \text{Bin}(2, p)$, where $p \in \{\frac{1}{2}, \frac{1}{3}\}$.
- Problem boils down to estimate the value of p .

Example (cont.)

Now consider the following table, where the entries are $P_p(X = x)$ for each possible values of x and p .

	$x = 0$	$x = 1$	$x = 2$
$p = 1/2$	1/4	1/2	1/4
$p = 1/3$	4/9	4/9	1/9

From first column, we see that for $x = 0$, the $P(X = 0)$ is maximum if $p = 1/3$. Hence if we observe $x = 0$ (that is no black balls in the sample), it is plausible to take $p = 1/3$ and the MLE of p is $1/3$.

From second column, we see that for $x = 1$, the $P(X = 1)$ is maximum if $p = 1/2$.

From third column, we see that for $x = 2$, the $P(X = 2)$ is maximum if $p = 1/2$.

Example (cont.)

Hence, the maximum likelihood estimator of p is

$$\hat{p} = \begin{cases} \frac{1}{3} & \text{if } x = 0 \\ \frac{1}{2} & \text{if } x = 1, 2. \end{cases}$$

Remark 12: If $x = 0$ occur, it is more likely that there are lesser number of black balls and hence the estimate turns out to be 1:2. For other values of x , it is 1:1.

Definition 6: Let $\mathbf{X} = (X_1, \dots, X_n)$ be a RS from a population with PMF/PDF $f(x; \boldsymbol{\theta})$. The function

$$L(\boldsymbol{\theta}, \mathbf{x}) = f_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{i=1}^n f(x_i, \boldsymbol{\theta})$$

considered as a function of $\boldsymbol{\theta} \in \Theta$ for any fixed $\mathbf{x} \in \mathcal{X}$ ($\mathcal{X}$ is support of the RS), is called the likelihood function.

Definition 7: For a sample point $\mathbf{x} \in \mathcal{X}$, let $\hat{\boldsymbol{\theta}}(\mathbf{x})$ be a value in Θ at which $L(\boldsymbol{\theta}, \mathbf{x})$ attains its maximum as a function of $\boldsymbol{\theta}$, with $\mathbf{x}$ held fixed. Then maximum likelihood estimator of the parameter $\boldsymbol{\theta}$ based on a RS $\mathbf{X}$ is $\hat{\boldsymbol{\theta}}(\mathbf{X})$.

Remark 13:

- MLE always lies in the parametric space.
- Problem of finding MLE boils down to finding maxima of a function, the likelihood function.
- Sometimes it is easier to work with $l(\boldsymbol{\theta}, \boldsymbol{x}) = \ln L(\boldsymbol{\theta}, \boldsymbol{x})$ than $L(\boldsymbol{\theta}, \boldsymbol{x})$. Note that $\ln(\cdot)$ is a strictly increasing function on the positive side of $\mathbb{R}$. However, it does not work in many examples.

Examples

Example 9: $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} P(\lambda)$, $\lambda > 0$. For $\lambda > 0$, the log-likelihood function is

$$l(\lambda, \mathbf{x}) = \ln L(\lambda, \mathbf{x}) = -n\lambda + n\bar{x} \ln \lambda - \sum_{i=1}^n \ln(x!).$$

$\frac{dl}{d\lambda} = 0 \implies \lambda = \bar{x}$. Also $\frac{d^2l}{d\lambda^2} < 0$ for all $\lambda > 0$. Hence $l(\lambda, \mathbf{x})$ maximizes at $\lambda = \bar{x}$ and the MLE of λ is $\hat{\lambda} = \bar{X}$.

Remark 14: Now onwards, we will write $L(\boldsymbol{\theta})$ instead of $L(\boldsymbol{\theta}, \mathbf{x})$. Similarly, we will use $l(\boldsymbol{\theta})$.

Example 10: $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, 1)$, $\mu \in \mathbb{R}$. The MLE of μ is $\hat{\mu} = \bar{X}$.

Examples

Example 11: $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$, where $\mu \in \mathbb{R}$ and $\sigma > 0$.
The log-likelihood function is

$$l(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Now $\frac{\partial l}{\partial \mu} = 0$ and $\frac{\partial l}{\partial \sigma^2} = 0$ implies that $\mu = \bar{x}$ and $\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$.
Find the Hessian matrix and show that it is negative definite. Hence
MLEs of μ and σ^2 are $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$, respectively.

Example 12: $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$, where $\sigma > 0$. MLEs of σ^2
is $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$.

Remark 15: Note that the estimator of σ^2 are different in the last two examples. It shows that the MLE may update itself based on any available information on the parameters.

Examples

Example 13: $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, 1)$, $\mu \leq 0$. For $\mu \leq 0$, the log-likelihood function is

$$l(\mu) = C - \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2,$$

where C is a constant (does not depend on unknown parameter).

Now, $\frac{dl}{d\mu} = n(\bar{x} - \mu)$. Clearly for $\bar{x} > 0$, $\frac{dl}{d\mu} = 0$ does not possess a solution in the parametric space. However, $\frac{dl}{d\mu} > 0$ for all $\mu \leq 0$. Hence, for $\bar{x} > 0$, $l(\mu)$ is an increasing function and it takes its maximum value at $\mu = 0$. On the other hand if $\bar{x} \leq 0$, $\frac{dl}{d\mu} = 0$ possesses a solution and it is $\mu = \bar{x}$. The second derivative condition can be easily checked. Hence, the MLE of μ is

$$\hat{\mu} = \begin{cases} \bar{X} & \text{if } \bar{X} \leq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Example 14: Let X_1 be a sample of size one from $Bernoulli(\frac{1}{1+e^\theta})$, where $\theta \geq 0$.

In this case $L(\theta, 0) = \frac{e^\theta}{1+e^\theta}$ and $L(\theta, 1) = \frac{1}{1+e^\theta}$. Clearly MLE does not exist for $x = 0$ as $L(\theta, 0)$ is an increasing function of θ . On the other hand MLE exist for $x = 1$ and it is $\hat{\theta} = 0$.

Remark 16:

- MLE may not exist.
- The log-likelihood is not used here.

Example 15: $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} U(0, \theta)$, $\theta > 0$. The likelihood function is

$$L(\theta) = \frac{1}{\theta^n} \text{ if } 0 < x_1, \dots, x_n \leq \theta.$$

Clearly $L(\theta)$ is a decreasing function on $\theta > \max\{x_1, x_2, \dots, x_n\} = x_{(n)}$ and it takes its maximum value at $\theta = x_{(n)}$. Hence the MLE of θ is $\hat{\theta} = X_{(n)}$.

Remark 17: When support depends on unknown parameters, you should take proper care to find MLE, if it exists.

Examples

Example 16: $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} U\left(\theta - \frac{1}{2}, \theta + \frac{1}{2}\right), \theta \in \mathbb{R}.$

The likelihood function is

$$\begin{aligned} L(\theta) &= 1 \text{ if } \theta - \frac{1}{2} \leq x_1, \dots, x_n \leq \theta + \frac{1}{2} \\ &= 1 \text{ if } x_{(n)} - \frac{1}{2} \leq \theta \leq x_{(1)} + \frac{1}{2}, \end{aligned}$$

where $x_{(n)} = \max\{x_1, \dots, x_n\}$ and $x_{(1)} = \min\{x_1, \dots, x_n\}$.

As $X_{(n)} - X_{(1)} \leq 1$ with probability one, $[x_{(n)} - \frac{1}{2}, x_{(1)} + \frac{1}{2}]$ is a non-empty interval. Also $L(\theta)$ maximizes at any point in the interval.

Hence any point in the interval $[X_{(n)} - \frac{1}{2}, X_{(1)} + \frac{1}{2}]$ is a MLE of θ .

In particular, a MLE of θ is $\hat{\theta} = \alpha \left(X_{(n)} - \frac{1}{2}\right) + (1 - \alpha) \left(X_{(1)} + \frac{1}{2}\right)$ for any value of $\alpha \in [0, 1]$.

Remark 18: This example shows that MLE may not be unique.

Invariance Property of MLE

Theorem 10: (Without Proof) If $\hat{\theta}$ is MLE of θ , then for any function $\tau(\cdot)$ defined on Θ , the MLE of $\tau(\theta)$ is $\tau(\hat{\theta})$.

Example 17: $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} P(\lambda)$, $\lambda > 0$. To find the MLE of $P(X_1 = 0)$, we can proceed as follows:

Note that $P(X_1 = 0) = e^{-\lambda}$ and we know that the MLE of λ is $\bar{X}$. Hence the MLE of $P(X_1 = 0)$ is $e^{-\bar{X}}$.

Remark 19:

- Note that invariance property may not hold for MME.
- The proof of above theorem is straight forward for a strictly monotone function $\tau(\cdot)$. However, the proof is little involved for a general function.

Comparison of Different Estimators

- We have considered two different methods of estimation.
- One may want to know which method provide a better estimator in a particular situation.
- Can we talk about average error? Can we talk about average squared error?
- There are some desirable properties of an estimator. Some of them are discussed here.

Unbiased Estimator

Definition 8: A statistic T is said to be an unbiased estimator (UE) of a parametric function $\tau(\theta)$ if $E_{\theta}(T) = \tau(\theta)$ for all $\theta \in \Theta$, the parametric space.

Remark 20:

- Unbiasedness tells us that there is no error on an average taken over all samples.
- Please note that **all** $\theta \in \Theta$ in the definition.
- An estimator which is not unbiased is called a biased.
- Bias of T as an estimator of $\tau(\theta)$ is defined by $Bias(T) = E_{\theta}(T) - \tau(\theta)$ for all $\theta \in \Theta$.
- In general, for unbiased estimator invariance property does not hold true.

Example

Example 18: Let $X_1, \dots, X_n$ be a RS from a population with mean $\mu \in \mathbb{R}$. Then $\bar{X}$ is an unbiased estimator for μ as $E_\mu(\bar{X}) = \mu$ for all $\mu \in \mathbb{R}$.

Example 19: $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} U(0, \theta)$, $\theta > 0$. We saw that the MLE of θ is $X_{(n)}$. Now we want to check if $X_{(n)}$ is unbiased or not. It is easy to see that the CDF of $X_{(n)}$ is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \left(\frac{x}{\theta}\right)^n & \text{if } 0 \leq x < \theta \\ 1 & \text{otherwise.} \end{cases}$$

Now, check that $E_\theta(X_{(n)}) = \frac{n}{n+1}\theta$ for all $\theta > 0$. Hence, $X_{(n)}$ is a biased estimator for θ and $\text{Bias}(X_{(n)}) = -\frac{1}{n+1}\theta$. It is very easy to see that $T = \frac{n+1}{n}X_{(n)}$ is an unbiased estimator of θ .

Examples

Example 20: Let $X_1, \dots, X_n$ be a RS from a population with mean $\mu \in \mathbb{R}$ and finite variance σ^2 . Define $T_1 = X_1$, $T_2 = \frac{1}{2}(X_1 + X_2) \dots, T_n = \bar{X}$. It is easy to verify that $E(T_i) = \mu$ for all $\mu \in \mathbb{R}$ and for all $i = 1, 2, \dots, n$.

Remark 21: There may be more than one unbiased estimator for a parametric function. Which one to prefer?

Example 21: Let X be distributed as $Bin(2, p)$ ($p \in (0, 1)$) and suppose that $\tau(p) = \frac{1}{p}$. We want to check if there is an UE for $\frac{1}{p}$. If there exists an UE, say $\delta(X)$ for $1/p$, then it will satisfy

$$E_p(\delta(X)) = \delta(0) \binom{2}{0} q^2 + \delta(1) \binom{2}{1} pq + \delta(2) \binom{2}{2} p^2 = \frac{1}{p},$$

for all p , where $q = 1 - p$. Now for $p \rightarrow 0$, the left side tends to $\delta(0)$ and the right side tends to ∞ . Hence UE for $1/p$ does not exist in this case.

Remark 22: Unbiased estimator may not exist for a parametric function.

Mean Square Error

Definition 9: Mean square error (MSE) of a statistic T as an estimator of θ is defined by $MSE(T) = E((T - \theta)^2)$.

Remark 23:

- If T is UE for θ , then $MSE(T) = Var(T)$.
- $MSE(T) = Var(T) + (Bias(T))^2$.
- An estimator with smaller value of MSE is preferred.

Example 22: Let $X_1, \dots, X_n$ be a RS from a population with mean $\mu \in \mathbb{R}$ and finite variance σ^2 . $T_1, T_2, \dots, T_n$ are UE for μ . Which one to prefer? For $i = 1, 2, \dots, n$

$$MSE(T_i) = Var(T_i) = \frac{\sigma^2}{i}.$$

Hence, T_n has smallest MSE among these estimator.

A Comparison

Example 23: Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$, $n > 1$. Is the MLE of σ^2 is UE?

Using the fact that $\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi_{n-1}^2$, $E(\widehat{\sigma^2}) = \frac{n-1}{n} \sigma^2$. Hence it is not UE for σ^2 .

However $\widetilde{\sigma^2} = \frac{n}{n-1} \widehat{\sigma^2} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is UE for σ^2 .

Using the variance of a χ^2 -distribution, $Var(\widehat{\sigma^2}) = \frac{2n-2}{n^2} \sigma^4$ and $Var(\widetilde{\sigma^2}) = \frac{2}{n-1} \sigma^4$.

Hence $MSE(\widehat{\sigma^2}) = \frac{2n-1}{n^2} \sigma^4$ and $MSE(\widetilde{\sigma^2}) = \frac{2}{n-1} \sigma^4$.

It can be checked that $MSE(\widehat{\sigma^2}) < MSE(\widetilde{\sigma^2})$.

Remark 24: This example shows that biased estimator may have lower MSE and hence may be preferred over an unbiased estimator.

Consistent Estimator

Definition 10: Let T_n be an estimator based on a RS of size n . The estimator T_n is said to be consistent for θ if the sequence of random variables $\{T_n : n \geq 1\}$ converges to θ in probability for all $\theta \in \Theta$, i.e., if for all $\varepsilon > 0$ and all $\theta \in \Theta$,

$$\lim_{n \rightarrow \infty} P(|T_n - \theta| \leq \varepsilon) = 1.$$

Remark 25: Consistency says us that for a sample with reasonably large size, T_n is close to the true value of parameter with high probability.

Example 24: Let $X_1, X_2, \dots, X_n$ be a RS from a population with mean $\mu \in \mathbb{R}$. Then using SLLN, $\bar{X}_n$ is a consistent estimator for μ .

Example 25: $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} U(0, \theta)$, $\theta > 0$. We saw that the MLE of θ is $X_{(n)}$. Now using the CDF of $X_{(n)}$, it can be shown that $X_{(n)}$ is a consistent estimator of θ .

Large Sample Properties of MLE

- We will discuss some properties of MLE when the sample size is reasonably large.
- These properties are quite useful when it is difficult to find the exact distribution of MLE or when MLE does not exist in close form.
- We will state two theorems without proof. However there is a set of assumptions under which the theorems hold. We will first state these assumptions and then theorems.

Large Sample Properties of MLE (Contd.)

Let $X_1, X_2, \dots$ be a sequence of *i.i.d.* RVs from the population having PMF/PDF $f(x; \theta)$, where $\theta \in \Theta \subseteq \mathbb{R}$. Let the true value of θ is θ_0 .

Consider the following assumptions.

1. $\frac{\partial}{\partial \theta} \ln f(x; \theta)$, $\frac{\partial^2}{\partial \theta^2} \ln f(x; \theta)$, $\frac{\partial^3}{\partial \theta^3} \ln f(x; \theta)$ are finite for all $x \in \mathbb{R}$ and for all $\theta \in \Theta$.
2. $\int_{-\infty}^{+\infty} \frac{\partial}{\partial \theta} f(x; \theta) dx = 0$, $\int_{-\infty}^{+\infty} \frac{\partial^2}{\partial \theta^2} f(x; \theta) dx = 0$, and $\int_{-\infty}^{+\infty} \left\{ \frac{\partial}{\partial \theta} f(x; \theta) \right\}^2 dx > 0$ for all $\theta \in \Theta$.
3. For all $\theta \in \Theta$, $\left| \frac{\partial^3}{\partial \theta^3} \ln f(x; \theta) \right| < a(x)$, where $E(a(X_1)) < b$ for a constant b which is independent of θ .

Large Sample Properties of MLE (Contd.)

Theorem 11: Under these three assumptions, the likelihood equation has solution denoted by $\hat{\theta}_n(\mathbf{x})$, such that $\hat{\theta}_n(\mathbf{X})$ is consistent estimator for θ .

Theorem 12: Under these three assumptions,

$$\sqrt{nI(\theta)} \left(\hat{\theta}_n(\mathbf{X}) - \theta \right) \xrightarrow{\mathcal{L}} Z \sim N(0, 1),$$

i.e. for all $a \in \mathbb{R}$

$$\lim_{n \rightarrow \infty} P \left(\sqrt{nI(\theta)} \left(\hat{\theta}_n(\mathbf{X}) - \theta \right) \leq a \right) \rightarrow \Phi(a) = \int_{-\infty}^a \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt,$$

where $I(\theta) = -E \left(\frac{\partial^2}{\partial \theta^2} \ln f(X; \theta) \right)$ is called Fisher information.

Examples

Example 26: Let $X_1, X_2, \dots \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$. The MLE of p based on a sample of size n is $\hat{p}_n = \bar{X}_n$ and $I(p) = \frac{1}{p(1-p)}$. Using above theorems $\hat{p}_n$ is consistent for p and $\sqrt{n}(\hat{p}_n - p) \xrightarrow{\mathcal{L}} N(0, p(1-p))$.

Example 27: Let $X_1, X_2, \dots \stackrel{i.i.d.}{\sim} P(\lambda)$. The MLE of λ based on a sample of size n is $\hat{\lambda}_n = \bar{X}_n$ and $I(\lambda) = \frac{1}{\lambda}$. Using above theorems $\hat{\lambda}_n$ is consistent for λ and $\sqrt{n}(\hat{\lambda}_n - \lambda) \xrightarrow{\mathcal{L}} N(0, \lambda)$.

Remark 26:

- You can check that all the assumptions are hold true for last two examples.
- These two results can be proved using SLLN and CLT.

Example 28: Let $X_1, X_2, \dots \stackrel{i.i.d.}{\sim} U(0, \theta)$. The MLE of θ based on a sample of size n is $\hat{\theta}_n = X_{(n)}$. However, the first condition of assumption 2 does not hold. Hence we can not use previous theorems here. However we have already discussed that $X_{(n)}$ is consistent for θ . One can show that $n(\theta - X_{(n)}) \xrightarrow{\mathcal{L}} \text{Exp}(\theta)$. To show it find the CDF of $n(\theta - X_{(n)})$ and then use the definition of convergence in distribution.

Interval Estimation

Interval Estimation

- Our aim is to find an interval in $\Theta \subseteq \mathbb{R}$ such that the interval covers the unknown parameter with a specified probability.
- Note that for a RV X and two real constants $a \neq 0, b$,

$$P(a < X < b) = P(X < b < bX/a).$$

Though these two probabilities are same, there is a basic difference in RHS and LHS statements. For the LHS, we are taking about probability that a random quantity X belongs to a fixed interval (a, b) . For the RHS, we are taking about probability that a random interval $(X, bX/a)$ contains a fixed point b .

- For example, let $X \sim U(0, 1)$, $a = 0.5$, and $b = 1$. In this case $P(X < 1 < 2X) = P(0.5 < X < 1) = 0.5$.

Interval Estimation (Contd.)

- In interval estimation we will try to find a random interval based on the RS so that it contains the fixed unknown parameter with a pre-specified probability.
- Interval estimation is quite useful in practice. For example one may interested to find a upper limit of mean of toxic level of some drug or food.

Interval Estimation (Contd.)

Definition 11: An interval estimate of a real valued parameter θ is any pair of functions $L(x)$ and $U(x)$ of random sample only (do not involve any unknown parameters) that satisfy $L(x) \leq U(x)$ for all x in the support of RS. The random interval $[L(\mathbf{X}), U(\mathbf{X})]$ is called an interval estimator of θ .

Remark 27:

- Though in the definition the closed interval $[L(\mathbf{X}), U(\mathbf{X})]$ is considered, the interval may be closed, open or semi-open.
- If $L(x) = -\infty$, then $U(x)$ provides an upper limit. Similarly, if $U(x) = \infty$, then $L(x)$ provides a lower limit.

Example

Example 29: Let $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, 1)$. Consider $L_1(\mathbf{x}) = x_1 - 1$, $U_1(\mathbf{x}) = x_1 + 1$, $L_2(\mathbf{x}) = \bar{x} - 1$, and $U_2(\mathbf{x}) = \bar{x} + 1$. Then both $[L_1(\mathbf{X}), U_1(\mathbf{X})]$ and $[L_2(\mathbf{X}), U_2(\mathbf{X})]$ are interval estimator of μ .

Remark 28: Clearly we are loosing precision in interval estimation compared to point estimation. Do we have any gain? Consider the previous example. A reasonable point estimator of μ is $\bar{X}$. However, $P(\bar{X} = \mu) = 0$ as $\bar{X}$ is a CRV. On the other hand $P(\bar{X} - 1 \leq \mu \leq \bar{X} + 1) > 0$. Hence in interval estimation we have some confidence which we gain by reducing precision.

Confidence Interval (CI)

Definition 12: Let $\alpha \in (0, 1)$. An interval estimator $[L(\mathbf{X}), U(\mathbf{X})]$ is said to be a confidence interval of level $1 - \alpha$ (or a $100(1 - \alpha)\%$ confidence interval) if $P_{\theta}(L(\mathbf{X}) \leq \theta \leq U(\mathbf{X})) \geq 1 - \alpha$ for all $\theta \in \Theta$.

Remark 29:

- Typical values of α are 0.1, 0.05, 0.01.
- If we are able to have large number of realizations of a RS, and calculate a $100(1 - \alpha)\%$ CI for each realization, then about $100(1 - \alpha)\%$ times the true parameter will be inside the calculated CI. This is the physical mean of a CI and is called relative frequency interpretation of CI.

Method of Finding CI

Definition 13: A random variable $T = T(\mathbf{X}; \theta)$ is called a pivot (or a pivotal quantity) if the distribution of T does not involve any unknown parameters.

Example 30: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$ and μ and σ both are unknown. Then $\bar{X} - \mu$ is not a pivot as $\bar{X} - \mu \sim N(0, \sigma^2/n)$. However, $\frac{\sqrt{n}}{\sigma} (\bar{X} - \mu) \sim N(0, 1)$ and $\frac{\sqrt{n}}{S} (\bar{X} - \mu) \sim t_{n-1}$ and hence are pivot.

Example 31: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} Exp(\lambda)$. Then $2\lambda \sum_{i=1}^n X_i \sim \chi_{2n}^2$ (why?) and hence is a pivot.

Remark 30: Pivot is a function of random sample and unknown parameters, but its' distribution is independent of all unknown parameters. Hence pivot is not a statistic.

Method of Finding CI (Contd.)

- Let T be a pivot.
- Find two real numbers a and b such that

$$P(a \leq T(\mathbf{X}; \theta) \leq b) \geq 1 - \alpha.$$

Note that a and b are independent of all unknown parameters.

- A $100(1 - \alpha)\%$ CI for θ is

$$C(\mathbf{x}) = \{\theta \in \Theta : a \leq T(\mathbf{x}; \theta) \leq b\}.$$

Note that $C(\mathbf{x})$ does not involve any unknown parameters as a and b are independent of all unknown parameters.

Remark 31: If $T(\mathbf{x}; \theta)$ is monotone in $\theta \in \Theta$ for each $\mathbf{x}$, then $C(\mathbf{x})$ is an interval. Otherwise it will be a general set.

Examples

Example 32: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$, where $\mu \in \mathbb{R}$ is unknown and $\sigma > 0$ is known. We are interested in μ .

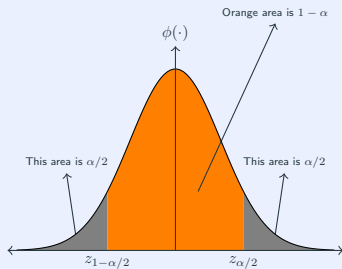
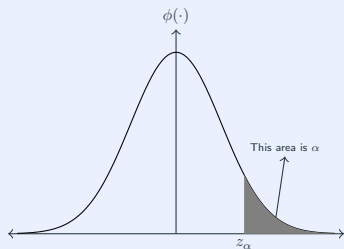
A pivot is $\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1)$. Let z_α be a real number such that $P(Z > z_\alpha) = \alpha$, where $Z \sim N(0, 1)$. We can take $a = z_{1-\alpha/2} = -z_{\alpha/2}$ (as $N(0, 1)$ distribution is symmetric about zero) and $b = z_{\alpha/2}$ (see next slide). Now

$$P\left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \leq z_{\alpha/2}\right) = 1 - \alpha$$
$$P\left(\bar{X} - \frac{\sigma}{\sqrt{n}}z_{\alpha/2} \leq \mu \leq \bar{X} + \frac{\sigma}{\sqrt{n}}z_{\alpha/2}\right) = 1 - \alpha.$$

Hence a $100(1 - \alpha)\%$ symmetric CI for μ is

$$C(\mathbf{X}) = \left[\bar{X} - \frac{\sigma}{\sqrt{n}}z_{\alpha/2}, \bar{X} + \frac{\sigma}{\sqrt{n}}z_{\alpha/2} \right].$$

Normal Curve



Remark 32:

- This is called symmetric CI as we have taken the middle part of the probability distribution.
- We can have infinite number of choices for a and b . For example $a = z_{1-\alpha_1}$ and $b = z_{\alpha_2}$, where $\alpha_1 > 0$, $\alpha_2 > 0$, and $\alpha_1 + \alpha_2 = \alpha$. Of course they will not be symmetric CI.
- The CI given in the previous slide is the minimum length CI based on $\bar{X}$.
- z_α is called upper α -point of standard normal distribution.

Examples

Example 33: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$, where $\mu \in \mathbb{R}$ is known and $\sigma > 0$ is unknown. We are interested in CI of σ^2 .

A pivot is $\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \sim \chi_n^2$. Let $\chi_{n,\alpha}^2$ be a real number such that $P(Z > \chi_{n,\alpha}^2) = \alpha$, where $Z \sim \chi_n^2$. We can take $a = \chi_{n,1-\alpha/2}^2$ and $b = \chi_{n,\alpha/2}^2$. Hence a $100(1 - \alpha)\%$ symmetric CI for σ^2 is

$$c(\mathbf{X}) = \left[\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{n,\alpha/2}^2}, \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{n,1-\alpha/2}^2} \right].$$

Remark 33: $\chi_{n,\alpha}^2$ is called upper α -point of χ^2 -distribution with degrees of freedom n .

Examples

Example 34: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$, where $\mu \in \mathbb{R}$ and $\sigma > 0$ are unknown. We are interested in CI of μ .

A pivot is $\frac{\sqrt{n}(\bar{X}-\mu)}{S} \sim t_{n-1}$, where $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$. Let $t_{n,\alpha}$ be a real number such that $P(Z > t_{n,\alpha}) = \alpha$, where $Z \sim t_n$. We can take $a = t_{n-1,1-\alpha/2} = -t_{n-1,\alpha/2}$ (as t -distribution is symmetric about zero) and $b = t_{n-1,\alpha/2}$. Hence a $100(1 - \alpha)\%$ symmetric CI for μ is

$$c(\mathbf{X}) = \left[\bar{X} - \frac{S}{\sqrt{n}} t_{n-1,\alpha/2}, \bar{X} + \frac{S}{\sqrt{n}} t_{n-1,\alpha/2} \right].$$

Remark 34: $t_{n,\alpha}$ is called upper α -point of t -distribution with degrees of freedom n .

Example 35: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$, where $\mu \in \mathbb{R}$ and $\sigma > 0$ are unknown. We are interested in CI of σ^2 .

A pivot is $\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi_{n-1}^2$. We can take $a = \chi_{n-1, 1-\alpha/2}^2$ and $b = \chi_{n-1, \alpha/2}^2$. Hence a $100(1 - \alpha)\%$ symmetric CI for σ^2 is

$$c(\mathbf{X}) = \left[\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\chi_{n-1, \alpha/2}^2}, \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\chi_{n-1, 1-\alpha/2}^2} \right].$$

- In many cases it is very difficult to find pivot for a small sample. For example it is difficult to find a pivot to construct CI for successes probability of a Bernoulli distribution.
- However, we may be able to find CI quite easily if the sample size is sufficiently large. This CI is called asymptotic confidence interval.
- We will use convergence in distribution (mainly CLT or large sample distribution of MLE) and in probability (consistent estimator).

Distribution Free Population Mean

- Let $X_1, X_2, \dots$ be i.i.d. random variables with mean μ and finite variance σ^2 . Then using CLT

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{\mathcal{L}} Z \sim N(0, 1).$$

- Now if n , sample size, is large, we can approximate the distribution of $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$ using a standard normal distribution. Hence

$$P\left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq z_{\alpha/2}\right) \approx 1 - \alpha.$$

- If σ is known and n is sufficiently large, we can use the last statement to find an asymptotic CI for μ and it is given by

$$\left[\bar{X}_n - \frac{\sigma}{\sqrt{n}}z_{\alpha/2}, \bar{X}_n + \frac{\sigma}{\sqrt{n}}z_{\alpha/2}\right].$$

Distribution Free Population Mean (Contd.)

If σ is unknown, we can proceed as follows:

- We know that $\frac{S_n}{\sigma} \xrightarrow{P} 1$, and hence

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \xrightarrow{\mathcal{L}} Z \sim N(0, 1).$$

- Hence $P\left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \leq z_{\alpha/2}\right) \approx 1 - \alpha$.
- An asymptotic CI for μ is given by

$$\left[\bar{X}_n - \frac{S_n}{\sqrt{n}} z_{\alpha/2}, \bar{X}_n + \frac{S_n}{\sqrt{n}} z_{\alpha/2} \right].$$

- Note that this method can be used for any distribution of $X_1, X_2, \dots, X_n$, as long as the conditions of CLT hold true. Hence it is called distribution free.

- Let $\hat{\theta}_n$ be a consistent estimator of θ and $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{\mathcal{L}} N(0, b^2(\theta))$, where $b(\theta) > 0$ for all $\theta \in \Theta$.
- Assume that $b(\cdot)$ is a continuous function.
- Then $\frac{b(\hat{\theta}_n)}{b(\theta)} \xrightarrow{P} 1$ and hence $\frac{\sqrt{n}(\hat{\theta}_n - \theta)}{b(\hat{\theta}_n)} \xrightarrow{\mathcal{L}} N(0, 1)$.
- A $100(1 - \alpha)\%$ asymptotic CI for θ is given by

$$\left[\hat{\theta}_n - \frac{b(\hat{\theta}_n)}{\sqrt{n}} z_{\alpha/2}, \hat{\theta}_n + \frac{b(\hat{\theta}_n)}{\sqrt{n}} z_{\alpha/2} \right].$$

- Under some regularity conditions, we may use MLE of θ in place of $\hat{\theta}_n$.

Examples

Example 36: $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$, where $p \in (0, 1)$.

We are interested to construct asymptotic CI for p .

We know that $\hat{p}_n = \bar{X}_n \xrightarrow{P} p$ and $\frac{\sqrt{n}(\bar{X}_n - p)}{\sqrt{p(1-p)}} \xrightarrow{\mathcal{L}} N(0, 1)$.

Here $b(p) = \sqrt{p(1-p)}$, which is a continuous function in $p \in (0, 1)$.

Hence $\frac{\sqrt{n}(\bar{X}_n - p)}{\sqrt{\bar{X}_n(1-\bar{X}_n)}} \xrightarrow{\mathcal{L}} N(0, 1)$.

A $100(1 - \alpha)\%$ asymptotic CI for p is

$$\left[\bar{X}_n - \sqrt{\frac{\bar{X}_n(1 - \bar{X}_n)}{n}} z_{\alpha/2}, \bar{X}_n + \sqrt{\frac{\bar{X}_n(1 - \bar{X}_n)}{n}} z_{\alpha/2} \right].$$

Hypothesis Testing

Testing of Hypothesis

- We have discussed point and interval estimation, where we try to find meaningful guess for unknown parameters.
- In testing of hypothesis, we do not guess the value. We try to check if some statement regarding parameters is true or not.

Example: Coin Toss

Example 37: A coin is tossed 80 times, and head are obtained 55 times. Can we conclude that the coin is significantly fair?

- Here $n = 80$, $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$.
- We want to test $p = 0.5$ or $p \neq 0.5$.
- $\bar{X} = 55/80 = 0.6875$.
- If p is actually equal to 0.5, using CLT we have

$$T_n = \frac{\sqrt{n} (\bar{X}_n - 0.5)}{\sqrt{0.5 \times (1 - 0.5)}} \approx N(0, 1).$$

- The observed value of $T_n = 3.3541$.
- Conclusion: It seems quite reasonable to reject the hypothesis $p = 0.5$, as the observed value of T_n is too extreme with respect to a standard normal distribution.

Example: Coin Toss (Contd.)

Example 38: A coin is tossed 80 times, and head are obtained 35 times. Can we conclude that the coin is significantly fair?

- Here the observed value of $T_n = -1.1180$.
- Conclusion: Data do not suggest to reject the fact that the coin is fair, as the observed value of T_n is not extreme with respect to a standard normal distribution.
- Note that in the last two examples we have talked about extreme or not extreme. The question is: Which values are considered as extreme and which are not?
- More precisely, we are rejecting $p = 0.5$ if the observation belong to the set $\{x : |T_n| > C\}$. What value of C should we choose?

Some Definitions

Definition 14: A hypothesis is a statement about the unknown parameter(s).

Definition 15: Suppose that one wants to choose between two reasonable hypotheses $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$, where $\Theta_0 \subset \Theta$, $\Theta_1 \subset \Theta$ and $\Theta_0 \cap \Theta_1 = \emptyset$. We call H_0 and H_1 are null hypothesis and alternative hypothesis, respectively.

Remark 35: The aim here is to choose one hypothesis among null and alternative hypotheses. As the roles of these two hypotheses are asymmetric, we need to be careful about labeling null and alternative hypotheses.

Approach: As illustrated in the examples, we will consider a reasonable statistic and make the choice based on the statistic.

Some Definitions (Contd.)

Definition 16: Let R be a subset of $\mathbb{R}^n$ such that we reject H_0 if $x \in R$. Then R is called rejection region or critical region. R^c is called acceptance region.

Definition 17: The error committed by rejecting H_0 when it is actually true is called Type-I Error. Error committed by accepting H_0 when it is actually false is called Type-II Error.

	H_0 true	H_1 true
Accept H_0	✓	Type-II Error
Reject H_0	Type-I Error	✓

Aim: To choose R such that probabilities of errors are as small as possible.

Example

Example 39: Let $X_1, X_2, \dots, X_9 \stackrel{i.i.d.}{\sim} N(\theta, 1)$. Suppose that we are want to test $H_0 : \theta = 5.5$ against $H_1 : \theta = 7.5$. Let use consider two critical regions $R_1 = \{\mathbf{x} \in \mathbb{R}^9 : \bar{x} > 6\}$ and $R_2 = \{\mathbf{x} \in \mathbb{R}^9 : \bar{x} > 7\}$. Let us compute the probability of errors. For R_1 ,

$$P(\text{Type-I Error}) = P_{\theta=5.5}(\bar{X} > 6) = 1 - \Phi(3(6 - 5.5)) = 0.06681.$$

$$P(\text{Type-II Error}) = P_{\theta=7.5}(\bar{X} \leq 6) = \Phi(3(6 - 7.5)) \sim 0.$$

Similarly the probabilities for R_2 can be computed and given in following table.

	R_1	R_2
P(Type-I)	0.06681	0
P(Type-II)	0	0.06681

Remark 36:

- Note that in the previous example $R_2 \subset R_1$.
- If we take $R = \emptyset$, then $P(\text{Type-I error}) = 0$ and $P(\text{Type-II error}) = 1$.
- If we take $R = \mathbb{R}^n$, then $P(\text{Type-I error}) = 1$ and $P(\text{Type-II error}) = 0$.
- If we try to reduce probability of one error, probability of the other one increases.
- For hypothesis testing the approach is as follows: Put a bound on the probability of Type-I error and try to minimize the probability of Type-II error.

Definition 18: The power function of a critical region, denoted by $\beta : \Theta_1 \cup \Theta_0 \rightarrow [0, 1]$, is the probability of rejecting the null hypothesis H_0 when θ is the true value of the parameter, i.e.,

$$\beta(\theta) = P_{\theta}(\mathbf{X} \in R).$$

Remark 37:

- For $\theta \in \Theta_0$, $\beta(\cdot)$ is the probability of Type-I error.
- For $\theta \in \Theta_1$, $\beta(\cdot)$ is one minus probability of Type-II error.

Example 40: Let $X_1, X_2, \dots, X_9 \stackrel{i.i.d.}{\sim} N(\theta, 1)$. Suppose that we are want to test $H_0 : \theta = 5.5$ against $H_1 : \theta = 7.5$. Let us consider the critical region $R_2 = \{\mathbf{x} \in \mathbb{R}^9 : \bar{x} > 7\}$. The power function is defined by

$$\beta(\theta) = 1 - \Phi(21 - 3\theta) \text{ for } \theta = 5.5, 7.5.$$

Definition 19: Let $\alpha \in (0, 1)$ be a fixed real number. A test for $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$ with power function $\beta(\cdot)$ is called a size α test if $\sup_{\theta \in \Theta} \beta(\theta) = \alpha$.

Definition 20: A test is called level α if $\beta(\theta) \leq \alpha$ for all $\theta \in \Theta_0$.

Remark 38:

- Size of a test can be considered as worst possible probability of Type-I error.
- If a test is of size α , then it is of level α .

Test Function

Definition 21: A function $\psi : \mathcal{X}^n \rightarrow [0, 1]$ is called a critical function or test function, where $\psi(\mathbf{x})$ stands for the probability of rejecting H_0 when $\mathbf{X} = \mathbf{x}$ is observed. Here $\mathcal{X}^n$ is the sample space of the random sample of size n .

Definition 22: The power function of a test function is defined by $\beta(\theta) = E_\theta(\psi(\mathbf{X}))$ for all $\theta \in \Theta_0 \cup \Theta_1$.

Example 41: Let $X_1, X_2, \dots, X_9 \stackrel{i.i.d.}{\sim} N(\theta, 1)$. Suppose that we are want to test $H_0 : \theta = 5.5$ against $H_1 : \theta = 7.5$. Let us consider two critical regions $R_1 = \{\mathbf{x} \in \mathbb{R}^9 : \bar{x} > 6\}$ and $R_2 = \{\mathbf{x} \in \mathbb{R}^9 : \bar{x} > 7\}$. The critical regions R_1 and R_2 , respectively, can be expressed as the test functions

$$\psi_1(\mathbf{x}) = \begin{cases} 1 & \text{if } \bar{x} > 6 \\ 0 & \text{if } \bar{x} \leq 6, \end{cases} \quad \text{and} \quad \psi_2(\mathbf{x}) = \begin{cases} 1 & \text{if } \bar{x} > 7 \\ 0 & \text{if } \bar{x} \leq 7. \end{cases}$$

Remark 39:

- Last example shows that test function is an alternative way of writing critical region.
- Then what does we gain by defining test function? To discuss this, first note that if $\psi(x)$ is a probability, then why should we restrict only 0 and 1? We can consider other values between 0 and 1. And this is the gain we have when dealing with discrete distributions.

Most Powerful Test

Definition 23: Consider the collection $\mathcal{C}$ of all level α tests for $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$. A test belonging to $\mathcal{C}$ with power function $\beta(\cdot)$ is called uniformly most powerful (UMP) level α test if $\beta(\theta) \geq \beta^*(\theta)$ for all $\theta \in \Theta_1$, where $\beta^*(\cdot)$ is the power function of any other test in $\mathcal{C}$. If the alternative hypothesis is simple (that means that Θ_1 is singleton), the test is called most powerful (MP) level α test.

Remark 40:

- MP test (or in some cases UMP test) can be found using Neyman-Pearson Lemma (will not covered in this course).
- UMP test may not exist in many practical cases.
- Therefore, we will discuss a procedure to construct a reasonable test. This test is called likelihood ratio test.

Likelihood Ratio Test: Algorithm

- We want to test $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \in \Theta_1$.
- Consider

$$\Lambda(\mathbf{x}) = \frac{\sup_{\theta \in \Theta_0} L(\theta, \mathbf{x})}{\sup_{\theta \in \Theta_0 \cup \Theta_1} L(\theta, \mathbf{x})}.$$

$\Lambda(\mathbf{X})$ is called likelihood ratio test statistic.

- Likelihood ratio level α test is given by

$$\psi(\mathbf{x}) = \begin{cases} 1 & \text{if } \Lambda(\mathbf{x}) < k \\ \gamma & \text{if } \Lambda(\mathbf{x}) = k \\ 0 & \text{if } \Lambda(\mathbf{x}) > k, \end{cases}$$

where γ and k are such that $E_{\theta}(\psi(\mathbf{X})) \leq \alpha$ for all $\theta \in \Theta_0$.

Likelihood Ratio Test: Discussion

- $\sup_{\theta \in \Theta_0} L(\theta, \mathbf{x})$ can be considered as the max value of the likelihood function (probability for DRVs) over Θ_0 when $\mathbf{X} = \mathbf{x}$ is observed.
- Similarly, $\sup_{\theta \in \Theta_0 \cup \Theta_1} L(\theta, \mathbf{x})$ can be considered as the max value of the likelihood function over $\Theta_0 \cup \Theta_1$ when $\mathbf{X} = \mathbf{x}$ is observed.
- Clearly $\Lambda(\mathbf{x}) \in [0, 1]$.
- We reject H_0 if Λ is small, as in this case the likelihood under Θ_0 is small compared to that over $\Theta_0 \cup \Theta_1$. This means that the observed values are more likely under Θ_1 than under Θ_0 . Hence we reject H_0 .

Example

Example 42: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$, where σ is known. Let μ_0 be a real number. We are interested to test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$.

Here $\Theta_0 = \{\mu_0\}$ and $\Theta_1 = \mathbb{R} \setminus \{\mu_0\}$. Hence $\Theta_0 \cup \Theta_1 = \mathbb{R}$. Now

$$\sup_{\mu \in \Theta_0} L(\mu) = L(\mu_0) = \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^n \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu_0)^2 \right],$$

and

$$\sup_{\mu \in \Theta_0 \cup \Theta_1} L(\mu) = L(\bar{x}) = \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^n \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2 \right].$$

Example (Contd.)

Hence

$$\Lambda = \exp \left[-\frac{n}{2\sigma^2} (\bar{x} - \mu_0)^2 \right] < k \iff |\bar{x} - \mu_0| > k_1.$$

Likelihood ratio level α test is given by

$$\psi(\mathbf{x}) = \begin{cases} 1 & \text{if } |\bar{x} - \mu_0| > k_1 \\ 0 & \text{otherwise,} \end{cases}$$

where k_1 is such that

$$\begin{aligned} E_{\mu_0}(\psi(\mathbf{X})) = \alpha &\iff P_{\mu_0} \left(\frac{\sqrt{n}}{\sigma} |\bar{X} - \mu_0| > \frac{\sqrt{n}}{\sigma} k_1 \right) = \alpha \\ &\iff P_{\mu_0} \left(\frac{\sqrt{n}}{\sigma} (\bar{X} - \mu_0) > \frac{\sqrt{n}}{\sigma} k_1 \right) = \frac{\alpha}{2} \\ &\iff \frac{\sqrt{n}}{\sigma} k_1 = z_{\alpha/2}. \end{aligned}$$

Example (Contd.)

Hence the likelihood ratio level α test is given by

$$\psi(\mathbf{x}) = \begin{cases} 1 & \text{if } \frac{\sqrt{n}|\bar{x} - \mu_0|}{\sigma} > z_{\alpha/2} \\ 0 & \text{otherwise.} \end{cases}$$

Remark 41: $\frac{\sqrt{n}|\bar{x} - \mu_0|}{\sigma}$ can be considered as standard distance between $\bar{x}$ and μ_0 . Hence this test rejects the H_0 if the distance is large, which is quite intuitive.

Example

Example 43: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$. Let μ_0 be a real number. We are interested to test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$.

Here $\Theta_0 = \{\mu_0\} \times \mathbb{R}^+$ and $\Theta_1 = \mathbb{R} \setminus \{\mu_0\} \times \mathbb{R}^+$. Hence $\Theta_0 \cup \Theta_1 = \mathbb{R} \times \mathbb{R}^+$. Now

$$\sup_{\Theta_0} L(\mu, \sigma^2) = \sup_{\sigma > 0} L(\mu_0, \sigma^2) = \left(\frac{2\pi e}{n} \sum_{i=1}^n (x_i - \mu_0)^2 \right)^{-n/2},$$

and

$$\sup_{\Theta_0 \cup \Theta_1} L(\mu, \sigma^2) = \left(\frac{2\pi e}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right)^{-n/2}.$$

Example (Contd.)

Hence

$$\Lambda = \left(\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\sum_{i=1}^n (x_i - \mu_0)^2} \right)^{n/2} = \left(1 + \frac{n(\bar{x} - \mu_0)^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)^{-n/2}.$$

Note that for $x > 0$, $f(x) = (1 + x)^{-n/2}$ is a decreasing function in x .

Hence

$$\Lambda < k \iff \frac{n(\bar{x} - \mu_0)^2}{\sum_{i=1}^n (x_i - \bar{x})^2} > k_1 \iff \frac{\sqrt{n}|\bar{x} - \mu_0|}{s} > k_2,$$

where $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$. The likelihood ratio test is given by

$$\psi(\mathbf{x}) = \begin{cases} 1 & \text{if } \frac{\sqrt{n}|\bar{x} - \mu_0|}{s} > k_2 \\ 0 & \text{otherwise,} \end{cases}$$

where $E_{\Theta_0}(\psi(\mathbf{X})) = \alpha$.

Example (Contd.)

Now we know that

$$\frac{\sqrt{n} |\bar{X} - \mu_0|}{S} \sim t_{n-1} \text{ under } H_0.$$

Hence $E_{\Theta_0}(\psi(\mathbf{X})) = \alpha \implies k_2 = t_{n-1; \alpha/2}$ and the likelihood ratio test is given by

$$\psi(\mathbf{x}) = \begin{cases} 1 & \text{if } \frac{\sqrt{n} |\bar{x} - \mu_0|}{s} > t_{n-1; \alpha/2} \\ 0 & \text{otherwise.} \end{cases}$$

Example

Example 44: Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$. Let σ_0 be a positive real number. We are interested to test $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 \neq \sigma_0^2$.

Here $\Theta_0 = \mathbb{R} \times \{\sigma_0^2\}$ and $\Theta_1 = \mathbb{R} \times \mathbb{R}^+ \setminus \{\sigma_0^2\}$. Hence $\Theta_0 \cup \Theta_1 = \mathbb{R} \times \mathbb{R}^+$. Now

$$\sup_{\Theta_0} L(\mu, \sigma^2) = \sup_{\mu \in \mathbb{R}} L(\mu, \sigma_0^2) = (2\pi\sigma_0^2)^{-n/2} \exp \left[-\frac{1}{2\sigma_0^2} \sum_{i=1}^n (x_i - \bar{x})^2 \right]$$

and

$$\sup_{\Theta_0 \cup \Theta_1} L(\mu, \sigma^2) = \left(\frac{2\pi e}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right)^{-n/2}.$$

Example (Contd.)

Hence

$$\Lambda = \left(\frac{(n-1)s^2}{n\sigma_0^2} \right)^{n/2} \exp \left[\frac{n}{2} - \frac{(n-1)s^2}{2\sigma_0^2} \right].$$

Note that for $x > 0$, $f(x) = \left(\frac{x}{n}\right)^{n/2} e^{n/2-x/2}$ is a convex function in x with unique maxima at $x = n$. Hence

$$\Lambda < k \iff \frac{(n-1)s^2}{\sigma_0^2} < k_1 \text{ or } \frac{(n-1)s^2}{\sigma_0^2} > k_2,$$

where $k_1 < k_2$. The likelihood ratio test is given by

$$\psi(\mathbf{x}) = \begin{cases} 1 & \text{if } \frac{(n-1)s^2}{\sigma_0^2} < k_1 \text{ or } \frac{(n-1)s^2}{\sigma_0^2} > k_2 \\ 0 & \text{otherwise,} \end{cases}$$

where $E_{\Theta_0}(\psi(\mathbf{X})) = \alpha$.

Example (Contd.)

Now we know that $T = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2$ under H_0 .

Hence $E_{\Theta_0}(\psi(\mathbf{X})) = \alpha \implies P_{\Theta_0}(T < k_1) + P_{\Theta_0}(T > k_2) = \alpha$.

There are infinite number of choices of (k_1, k_2) . The most popular one is as follows:

$$\begin{aligned} P_{\Theta_0}(T < k_1) &= \frac{\alpha}{2} = P(T > k_2) \\ \implies k_1 &= \chi_{n-1; 1-\alpha/2}^2 \text{ and } k_2 = \chi_{n-1; \alpha/2}^2. \end{aligned}$$

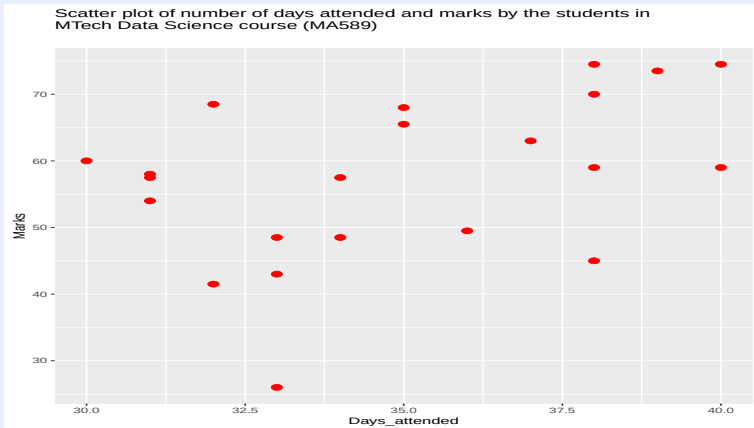
Hence the likelihood ratio test is given by

$$\psi(\mathbf{x}) = \begin{cases} 1 & \text{if } \frac{(n-1)s^2}{\sigma_0^2} < \chi_{n-1; 1-\alpha/2}^2 \text{ or } \frac{(n-1)s^2}{\sigma_0^2} > \chi_{n-1; \alpha/2}^2 \\ 0 & \text{otherwise.} \end{cases}$$

Simple Linear Regression

Research Question: What is the impact of attending classes on students' final marks

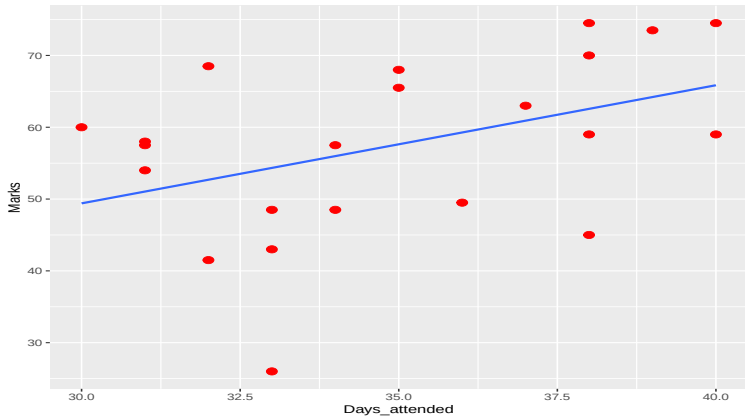
Let's start with a real data from IITG which you can feel about it!!



- We will do a regression analysis to the above data to find the answer. Is using regression appropriate here?
- **Strong Advice:** As a Data Scientist/Statistician, you have to **spent a lot of your time** and effort to **pre-process/clean the raw data** to make it **analysis ready**. It is part of your job to clean the data: so learn it quickly!!
- Here is the output after fitting a regression line in R:

Fitting Regression line

Scatter plot of number of days attended and marks by the students in MTech Data Science course (MA589):
with fitted regression line



Linear Models: SLR and MLR

We have one particular variable that we are interested in understanding or modeling, such as sales of a particular product, sale price of a home, or voting preference of a particular voter. This variable is called the **target, response, or dependent variable, and is usually represented** by y .

We have a set of p other variables that we think might be useful in predicting or modeling the target variable (the price of the product, the competitor's price, and so on; or the lot size, number of bedrooms, number of bathrooms of the home, and so on; or the gender, age, income, party membership of the voter, and so on). These are called the **predicting, or independent variables, and are usually represented** by $x_1, x_2, \dots, x_p$.

Our Aim

- To build a relationship between y and x .
- That means to write y as a function of x .
- Observed data: (x_i, y_i) for $i = 1, 2, \dots, n$
- We want a function of x , say $f(\cdot)$, such that

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

is as small as possible.

- There are infinite solutions for this problem.
- Therefore, we want to impose constraint the set of functions over which the minimization is performed.

- Typically, a regression analysis is used for one (or more) of three purposes:
 1. **modeling the relationship** between x and y ;
 2. **prediction of the target** variable (forecasting);
 3. and **testing of hypotheses**.

Linear Models: Meaning of Linearity

What does “Linear” mean?

A linear model is **linear in the parameters β** , but not necessarily in the **x 's**, e.g.

* $y = \beta_0 + \beta_1 x + \beta_2 x^2$ is a **linear model**

* $y = \beta_0 + \beta_1 x^{\beta_2}$ is a **non-linear model**

Transformation

Clearly $y = f(x) = \beta_0 x^{\beta_1}$ is **not a linear model**, but

$$\ln f(x) = \ln \beta_0 + \beta_1 \ln x$$

If we let $f^* = \ln f, \beta_0^* = \ln \beta_0, \beta_1^* = \beta_1, x^* = \ln x$, we have

$$f^* = \beta_0^* + \beta_1^* x^*,$$

which is a **linear model**.

Transformation

- Thus, although linear models seem to be simple and restrictive, they **can actually be quite flexible by transformation of the response and the predictors.**
- Linear models are **not just straight lines, they can be curved.**
Can you give an examples?

Transform the following **non-linear models to linear models:**

- $y = \frac{e^{\beta x}}{1+e^{\beta x}}$, where $y \in (0, 1)$.

- $y = \frac{1}{\beta_0 + \beta_1 x_1 + \beta_2 x_2}.$

Simple Linear Regression

- Just one predictor x , i.e. $p = 1$, where p indicates the number of predictors in a multiple linear regression model.
- The model for the **simple linear regression** is given by

$$y = \beta_0 + \beta_1 x + \epsilon,$$

where y is the **outcome** variable (random), x is the **independent/predictor** variable (non-random) and ϵ is the **random error term**. β_0 (intercept) and β_1 (slope) are model parameters (unknown constants).

- Equivalently, the **model can be written** for $i = 1, 2, \dots, n$ number of observations $(x_1, y_1), \dots, (x_n, y_n)$ as

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, i = 1, \dots, n.$$

- How do you **interpret** β_0 (intercept) and β_1 (slope)?

Assumptions:

1. The **expected value of the errors is zero**: $E(\epsilon_i) = 0$ for all i
2. The **variance of the error is constant** (homoscedasticity):
 $Var(\epsilon_i) = \sigma^2$ for all i
3. The **errors** (ϵ_i) are **uncorrelated** with each other: $Cov(\epsilon_i, \epsilon_j) = 0$
for all $i \neq j$

Assumptions: What they means?

The **errors** (ϵ_i) are **uncorrelated** with each other. This violation most often occurs in data that are ordered in time (time series data), where errors that are near each other in time are often similar to each other (such time-related correlation is called autocorrelation). **Violation of this assumption** can lead to very **misleading assessments** of the strength of the regression.

The **expected value of the errors is zero** ($E(\epsilon_i) = 0$ for all i). That is, it cannot be true that for certain observations the model is systematically too low, while for others it is systematically too high.

The assumption of **homoscedasticity (constant variance)**: each outcome observation (Y) has exact same variance (also for error). If Y follows Poisson distribution, this assumption is violated. Why?

Least Squares Estimation

- Goal: To **estimate** β_0, β_1 **by minimizing error** in some sense (e.g. squared error)
- One reasonable way is to use the **principle of Least Squares**, i.e. **minimize the objective function**

$$Q(\beta_0, \beta_1) = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

with respect to β_0, β_1 .

- **Differentiate** $Q(\beta_0, \beta_1)$ with respect to β_0, β_1 and **equate the partial derivatives to zero** to get the estimates $\hat{\beta}_0, \hat{\beta}_1$.
- The resulting equations are called **normal equations**:

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \quad \text{and} \quad \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0$$

Least Squares Estimation

- The **solution** is given by

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = S_{xy}/S_{xx}$$

where

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 \text{ and } S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

- $\hat{\beta}_0$ and $\hat{\beta}_1$ are called the **least squares estimator (LSE)** of β_0 and β_1 respectively.

Properties of $\hat{\beta}_0$ and $\hat{\beta}_1$

- $\hat{\beta}_0$ and $\hat{\beta}_1$ are linear combination of y_i 's.
- $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased for β_0 and β_1 , respectively.
- Variances of $\hat{\beta}_0$ and $\hat{\beta}_1$ are

$$Var(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$$

Def: [**Linear Estimator**] An estimator $\hat{\theta}$ is called linear estimator of θ if $\hat{\theta}$ is a **linear combination** of random observations.

Def: [**BLUE**] An estimator $\hat{\theta}$ is called the best linear unbiased estimator (**BLUE**) of θ if $\hat{\theta}$ is **linear and unbiased estimator** of θ and $\hat{\theta}$ has **minimum variance among** all linear unbiased estimator of θ .

Gauss-Markov Theorem (Without proof)

Theorem

If the model assumptions hold true, the LSEs $\hat{\beta}_0$ and $\hat{\beta}_1$ are BLUE of β_0 and β_1 , respectively.

A Few Definitions

- Fitted values: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, $i = 1, \dots, n$
- Residuals: Difference between observed and fitted values of y , i.e.

$$e_i = y_i - \hat{y}_i, \quad i = 1, \dots, n$$

- The objective function evaluated at the LSE, i.e. $Q(\hat{\beta}_0, \hat{\beta}_1)$ is called the residual sum of squares (SS_{Res}).

$$SS_{Res} = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2$$

Properties of Least Squares Fit:

1. $\sum_{i=1}^n e_i = 0$. (from the first normal equation)
2. $\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i$. (using (1))
3. $\sum_{i=1}^n x_i e_i = 0$. (from the second normal equation)
4. $\sum_{i=1}^n \hat{y}_i e_i = 0$. (using (1) and (3))

Estimation of Error Variance

- It can be shown that

$$E(SS_{Res}) = (n - 2)\sigma^2$$

- Hence, $\hat{\sigma}^2 = \frac{SS_{Res}}{n-2} = MS_{Res}$ is an unbiased estimator of σ^2 . Here MS_{Res} is residual mean square.
- Observed value of $\hat{\sigma}^2 = \frac{SS_{Res}}{n-2}$ is called **Residual variance**. It's **square root is called residual standard error**.
- Note that SS_{Res} depends on the model. Therefore, any violation of model assumptions has serious effect on $\hat{\sigma}^2$.

Hypothesis Testing

- We will talk about mainly following hypotheses:
 - $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$
 - $H_0 : \beta_0 = 0$ against $H_1 : \beta_0 \neq 0$
- Assumption: We need the following assumption along with previous assumptions.

$\epsilon = (\epsilon_1, \dots, \epsilon_n)$ has a n -dimensional normal distribution.

Hypothesis Testing: β_1

- Want to test the hypothesis that the **slope** parameter (β_1) equals to a constant (a value, say β_{10}):

$$H_0 : \beta_1 = \beta_{10} \text{ ag. } H_1 : \beta_1 \neq \beta_{10}$$

- Note that, $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$ and y_i 's are independent.
- $\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{S_{xx}}) \implies z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2/S_{xx}}} \sim N(0, 1)$. But σ is unknown.
- $\frac{(n-2)MS_{Res}}{\sigma^2} \sim \chi_{n-2}^2$. Also MS_{Res} and $\hat{\beta}_1$ are independent.
- Therefore, the test statistic is

$$t = \frac{\hat{\beta}_1 - \beta_{10}}{\sqrt{MS_{Res}/S_{xx}}} \sim t_{n-2}, \text{ under } H_0.$$

- Reject H_0 iff $|t| > t_{n-2, \alpha/2}$; (at level α).

Hypothesis Testing: β_0

- Want to test the hypothesis that the **intercept** parameter (β_0) equals to a constant (a value, say β_{00}):

$$H_0 : \beta_0 = \beta_{00} \text{ ag. } H_1 : \beta_0 \neq \beta_{00}$$

- $\hat{\beta}_0 \sim N(\beta_0, \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})) \implies z = \frac{\hat{\beta}_0 - \beta_0}{\sqrt{\sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}} \sim N(0, 1)$. But σ is unknown.
- $\frac{(n-2)MS_{Res}}{\sigma^2} \sim \chi_{n-2}^2$. Also MS_{Res} and $\hat{\beta}_1$ are independent.
- Therefore, the test statistic is

$$t = \frac{\hat{\beta}_0 - \beta_{00}}{\sqrt{MS_{Res}(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}})}} \sim t_{n-2}, \text{ under } H_0.$$

- Reject H_0 iff $|t| > t_{n-2, \alpha/2}$; (at level α).

Interval Estimation: β_0 and β_1

- To get the CI for β_0 and β_1 , the pivots are

$$\frac{\hat{\beta}_0 - \beta_0}{\sqrt{MS_{Res}\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}}, \text{ and } \frac{\hat{\beta}_1 - \beta_1}{\sqrt{MS_{Res}/S_{xx}}}, \text{ respectively.}$$

- A $100(1 - \alpha)\%$ CI for β_0 is

$$\left[\hat{\beta}_0 \pm t_{n-2, \alpha/2} \sqrt{MS_{Res} \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)} \right].$$

- A $100(1 - \alpha)\%$ CI for β_1 is

$$\left[\hat{\beta}_1 \pm t_{n-2, \alpha/2} \sqrt{\frac{MS_{Res}}{S_{xx}}} \right].$$

Interval Estimation: CI for σ^2

- To get the CI for σ^2 , the pivots is

$$\frac{(n-2)MS_{Res}}{\sigma^2} \sim \chi_{n-2}^2$$

- A $100(1 - \alpha)\%$ CI for σ^2 is

$$\left[\frac{(n-2)MS_{Res}}{\chi_{n-2;\alpha/2}^2}, \frac{(n-2)MS_{Res}}{\chi_{n-2;1-\alpha/2}^2} \right].$$

Interval Estimation: CI for mean response

- A regression model can be used to **estimate the mean response** $E(y)$ for a particular value of the regressor variable x . Let x_0 be a value of the regressor variable (must be with the range of original data on x). Then $E(y|x_0) = \beta_0 + \beta_1 x_0$.
- Then, $\hat{y}_0 = \widehat{E(y|x_0)} = \hat{\beta}_0 + \hat{\beta}_1 x_0$
- And, $\hat{y}_0 \sim N\left(\beta_0 + \beta_1 x_0, \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)\right)$
- Pivot: $\frac{\hat{y}_0 - (\beta_0 + \beta_1 x_0)}{\sqrt{MS_{Res} \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}} \sim t_{n-2}$
- A $100(1 - \alpha)\%$ CI for $\beta_0 + \beta_1 x_0$ is

$$\left[\hat{y}_0 \pm t_{n-2; \alpha/2} \sqrt{MS_{Res} \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} \right].$$

Prediction Interval for New Observation:

- Let x_0 be a value of the regressor variable
- The true value of the response is y_0 (corresponding to x_0).
- We want to provide an interval I such that $P(y_0 \in I) = 1 - \alpha$
- Note that the point estimate of y_0 is $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.
- Consider $\psi = y_0 - \hat{y}_0$.
- Then, $E(\psi) = 0, Var(\psi) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$
- $\psi \sim N \left(0, \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right) \right)$
- Pivot:
$$\frac{y_0 - \hat{y}_0}{\sqrt{MS_{Res} \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}} \sim t_{n-2}$$

Prediction Interval for New Observation:

- A $100(1 - \alpha)\%$ prediction interval is

$$\left[\hat{y}_0 \pm t_{n-2; \alpha/2} \sqrt{MS_{Res} \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)} \right].$$

Coefficient of determination: R^2

- Coefficient of determination:

$$R^2 = \frac{SS_{Reg}}{SS_T} = 1 - \frac{SS_{Res}}{SS_T} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2},$$

where $SS_{Reg} = \sum_i (\hat{y}_i - \bar{y})^2$

- It is a bounded quantity: $0 \leq R^2 \leq 1$
- R^2 is interpreted as the “**proportion of variation explained by the model**”.
- **Higher values of R^2 are desirable** (R^2 close to 1 indicates a good fit), but “how high is high” **depends on the context**.

Simple Linear Regression in Heights Data

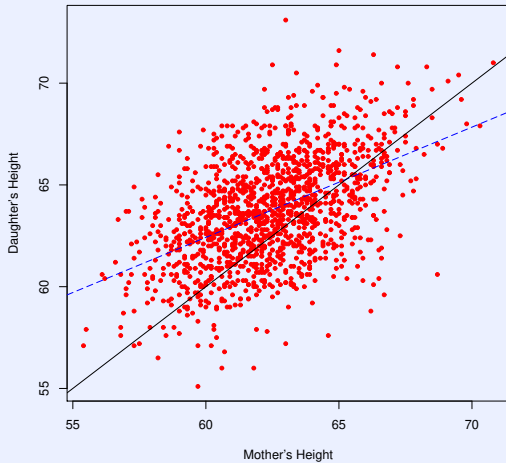
Heights Data Example

- Aim: To study inheritance of height from generation to generation (among females)
- Data on heights of $n = 1375$ mothers in the UK under the age of 65 and one of their adult daughters over the age of 18.
- Collected and organized during the period 1893–1898 by the famous statistician Karl Pearson.
- The dataset is publicly available
 - R-packages: alr4, UsingR
 - GitHub: https://github.com/kbroman/PearsonData/blob/master/mother_daughter.csv
- Sample correlation coefficient is 0.491

Simple Linear Regression: Using R

```
> height.lm <- lm(Dheight~Mheight,data=heights)
> summary(height.lm)
...
Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
(Intercept) 29.91744    1.62247   18.44  <2e-16 ***
Mheight      0.54175    0.02596   20.87  <2e-16 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1 1

Residual standard error: 2.266 on 1373 degrees of
freedom Multiple R-squared: 0.2408,
adjusted R-squared: 0.2402
F-statistic: 435.5 on 1 and 1373 DF,
p-value: < 2.2e-16
```



Caution

- Do you think a moderate or high absolute value of correlation coefficient makes a data ideal for LR?

Caution

- Do you think a moderate or high absolute value of correlation coefficient makes a data ideal for LR?
- Consider datasets given in next slide.

Anscombe's Examples (1973)

Dataset 1		Dataset 2		Dataset 3		Dataset 4	
x	y	x	y	x	y	x	y
10.00	8.04	10.00	9.14	10.00	7.46	8.00	6.58
8.00	6.95	8.00	8.14	8.00	6.77	8.00	5.76
13.00	7.58	13.00	8.74	13.00	12.74	8.00	7.71
9.00	8.81	9.00	8.77	9.00	7.11	8.00	8.84
11.00	8.33	11.00	9.26	11.00	7.81	8.00	8.47
14.00	9.96	14.00	8.10	14.00	8.84	8.00	7.04
6.00	7.24	6.00	6.13	6.00	6.08	8.00	5.25
4.00	4.26	4.00	3.10	4.00	5.39	19.00	12.50
12.00	10.84	12.00	9.13	12.00	8.15	8.00	5.56
7.00	4.82	7.00	7.26	7.00	6.42	8.00	7.91
5.00	5.68	5.00	4.74	5.00	5.73	8.00	6.89

Means & Standard Deviations

Data Set	N	<i>x</i>		<i>y</i>	
		Mean	SD	Mean	SD
1	11	7.50	2.03	9.00	3.32
2	11	7.50	2.03	9.00	3.32
3	11	7.50	2.03	9.00	3.32
4	11	7.50	2.03	9.00	3.32

The means and SDs for the 4 data sets are identical to two decimals.

Correlations between x and y

Data Set	Pearson r	R-squared	Adj. R-sq
1	0.82	0.67	0.63
2	0.82	0.67	0.63
3	0.82	0.67	0.63
4	0.82	0.67	0.63

Correlations, R-squared, adjusted R-squared, are all identical to two decimals.

ANOVA Summary Tables

Data Set	Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>p</i> -value
1	Regression	27.490	1	27.490	18.003	0.002
	Residual	13.742	9	1.527		
	Total	41.232	10			
2	Regression	27.470	1	27.470	17.972	0.002
	Residual	13.756	9	1.528		
	Total	41.226	10			
3	Regression	27.500	1	27.500	17.966	0.002
	Residual	13.776	9	1.531		
	Total	41.276	10			
4	Regression	27.510	1	27.510	17.990	0.002
	Residual	13.763	9	1.529		
	Total	41.273	10			

The Regression Coefficients

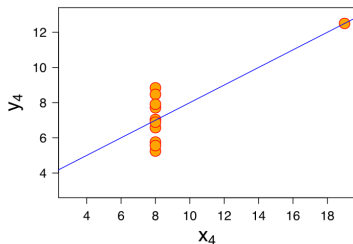
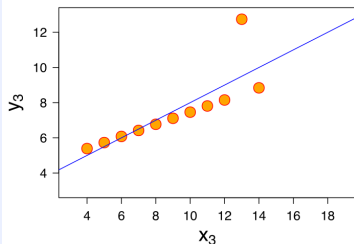
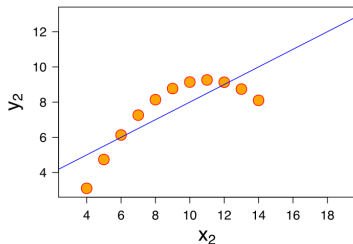
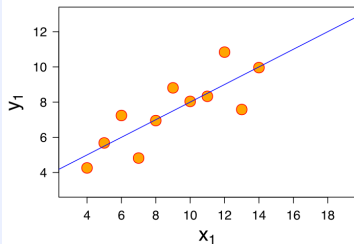
Data Set		Estimate	SE	t	p-value	95% CI	
						Lower	Upper
1	Constant	3.00	1.124	2.67	0.026	0.459	5.544
	x	0.50	0.118	4.24	0.002	0.233	0.766
2	Constant	3.00	1.124	2.67	0.026	0.459	5.546
	x	0.50	0.118	4.24	0.002	0.233	0.766
3	Constant	3.00	1.125	2.67	0.026	0.455	5.547
	x	0.50	0.118	4.24	0.002	0.233	0.767
4	Constant	3.00	1.125	2.67	0.026	0.456	5.544
	x	0.50	0.118	4.24	0.002	0.233	0.767

For all 4 models, $\hat{y} = 0.5x + 3$

Which dataset is best described by the model?

- Judging by everything we've just seen, it appears that the models are all equally good
- But if that were true, I wouldn't be doing this talk!
- Let's look at scatter plots between x and y

Importance of graphing data before analyzing it



Multiple Linear Regression

Multiple Linear Regression

- In general, the response (y) may be related to p regressors (input variables/predictors).
- The model

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i, i = 1, \dots, n$$

is called **multiple linear regression**. A regression model with p regressors.

- The parameters $\beta_j, j = 0, 1, 2, \dots, p$ are called regression coefficients.
- $\beta_j, j = 0, 1, 2, \dots, p$ represents the change in the average value of the response for a unit change in j^{th} regressor keeping other regressors fixed.

- Data (of sample size n): $(y_i, x_{i1}, x_{i2}, \dots, x_{ip}), i = 1, \dots, n$
- Scalar notation becomes cumbersome – so matrix notation is used
- Let

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, X = \begin{pmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{pmatrix}, \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}, \epsilon = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

- Then we can write the model in a more compact form:

$$y_{n \times 1} = X_{n \times (p+1)} \beta_{(p+1) \times 1} + \epsilon_{n \times 1}$$

- X is called the **design matrix**

Error Assumptions

- As before, ϵ_i 's satisfy
 - $E(\epsilon_i) = 0$ for $i = 1, \dots, n$
 - $Var(\epsilon_i) = \sigma^2$ for $i = 1, \dots, n$
 - $Cov(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$
- Equivalently, the assumptions can be written as
 - $E(\epsilon) = \mathbf{0}$
 - $Var(\epsilon) = \sigma^2 I_n$

Estimation of Regression Coefficients:

- Consider the objective function

$$\begin{aligned}Q(\beta) &= Q(\beta_0, \beta_1, \dots, \beta_p) \\&= \sum_{i=1}^n \left(y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip} \right)^2 \\&= (y - X\beta)^T (y - X\beta) \\&= (y^T y - 2\beta^T X^T y + \beta^T X^T X \beta)\end{aligned}$$

- The LSEs of $\beta_0, \beta_1, \dots, \beta_p$ are $\underset{\beta}{\operatorname{argmin}} Q(\beta)$.

Vector Derivative : Linear Function

- Let $\mathbf{x}$ be an $n \times 1$ column vector (the variable).
- Let $\mathbf{a}$ be a constant $n \times 1$ column vector.
- The derivative of a scalar-valued linear function with respect to the vector $\mathbf{x}$ is

$$\frac{\partial}{\partial \mathbf{x}} (\mathbf{a}^T \mathbf{x}) = \mathbf{a}.$$

Vector Derivative: Quadratic Function

- Let A be a constant $n \times n$ matrix.
- The derivative of a quadratic form with respect to the vector x is

$$\frac{\partial}{\partial x} (x^T A x) = (A + A^T) x.$$

- If A is a symmetric matrix, then the above formula simplifies to

$$\frac{\partial}{\partial x} (x^T A x) = 2Ax.$$

Estimation of Model Parameters:

- Differentiating $Q(\beta)$ with respect to β and setting the derivative to zero gives the following normal equations:

$$X^T X \beta = X^T y.$$

- The above system of linear equations in β is always consistent.
- If the matrix $X^T X$ is invertible (i.e. if X is of full rank), then the LSE is unique and given by

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

Fitted Value:

- The fitted value of response corresponding to regression $x = (1, x_1, \dots, x_p)$ is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$$

- Then $\hat{y} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)^T = X\hat{\beta} = X(X^T X)^{-1} X^T y = Hy$, where, $H = X(X^T X)^{-1} X^T$ is called **hat-matrix**.

- The residuals:

$$\mathbf{e} = \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} y_1 - \hat{y}_1 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix} = \mathbf{y} - \hat{\mathbf{y}} = (I - H)\mathbf{y}$$

Properties of LSE of β

- $\hat{\beta}$ is a linear function of y
- $\hat{\beta}$ is unbiased estimator of β . That is, $E(\hat{\beta}) = \beta$
- $Var(\hat{\beta}) = \sigma^2(X^T X)^{-1}$
- $\hat{\beta}$ is the BLUE of β .

Estimation of Error Variance (σ^2)

- It can be shown that

$$E(SS_{Res}) = (n - p - 1)\sigma^2$$

- Hence, $\hat{\sigma}^2 = \frac{SS_{Res}}{n-p-1} = MS_{Res}$ is an unbiased estimator of σ^2 . Here MS_{Res} is residual mean square.
- Observed value of $\hat{\sigma}^2 = \frac{SS_{Res}}{n-p-1}$ is called **Residual variance**. It's **square root is called residual standard error**.
- $\frac{(n-p-1)MS_{Res}}{\sigma^2} \sim \chi_{n-p-1}^2$.
- Here,

$$SS_{Res} = \sum_{i=1}^n e_i^2 = y^T y - \hat{\beta}_1^T X^T y,$$

Hypothesis Testing: Test for Significance of Regression

- Assumption: $\epsilon \sim N_n(0, \sigma^2 I_n)$.
- Want to test the hypothesis if there is a linear relationship between the response y and any of the regressors $x_1, \dots, x_n$:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$$

against

$$H_1 : \beta_j \neq 0 \text{ for atleast one } j = 1, \dots, p$$

- The test statistic is

$$F_0 = \frac{SS_{Reg}/p}{\hat{\sigma}^2} = \frac{SS_{Reg}/p}{SS_{Res}/(n-p-1)} \sim F_{p,n-p-1}, \text{ under } H_0.$$

- Reject H_0 iff $F_0 > F_{p,n-p-1;\alpha}$ (at level α)

Hypothesis Testing: Significance of regression coefficients

- Want to test

$$H_0 : \beta_j = 0 \text{ ag. } H_1 : \beta_j \neq 0$$

- Note that

$$\hat{\beta} \sim N_p \left(\beta, \sigma^2 (X^T X)^{-1} \right) \implies \hat{\beta}_j \sim N \left(\beta_j, \sigma^2 C_{jj} \right),$$

where C_{jj} is the j -th diagonal element of $(X^T X)^{-1}$

- The test statistic is

$$t_0 = \frac{\hat{\beta}_j}{\sqrt{MS_{Res} C_{jj}}} \sim t_{n-p-1} \text{ under } H_0$$

- Reject H_0 iff $|t_0| > t_{n-p-1, \alpha/2}$ (at level α)

Test of contribution of a subset of the regressors

-

$$y = X\beta + \epsilon \implies y = (X_1, X_2) \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \epsilon$$

- Want to test:

$$H_0 : \beta_2 = 0 \text{ ag. } H_1 : \beta_2 \neq 0$$

- Based on the full model ($y = X\beta + \epsilon$), $\hat{\beta} = (X^T X)^{-1} X^T y$ and $SS_{Reg}(\beta)$ has p degrees of freedom.
- To find the contribution of β_2 , fit the model assuming $\beta_2 = 0$.
- The reduced model is $y = X_1\beta_1 + \epsilon$, $\hat{\beta}_1 = (X_1^T X_1)^{-1} X_1^T y$ and $SS_{Reg}(\beta_1)$ has $p - r$ degrees of freedom. Where r denotes the number of components in β_2

Test of contribution of a subset of the regressors

- $SS_{Reg}(\beta_2 \mid \beta_1) = SS_{Reg}(\beta) - SS_{Reg}(\beta_1)$ can be used as a measure of contribution of β_2
- Note that if β_2 has significant contribution then $SS_{Reg}(\beta_2 \mid \beta_1)$ is large
- Therefore, the test statistic is

$$F_0 = \frac{SS_{Reg}(\beta_2 \mid \beta_1)/r}{\hat{\sigma}^2} = \frac{SS_{Reg}(\beta_2 \mid \beta_1)/r}{SS_{Res}/(n-p-1)} \sim F_{r,n-p-1}, \text{ under } H_0$$

- Reject H_0 iff $F_0 > F_{r,n-p-1;\alpha}$ (at level α)

Confidence intervals for β_j :

- Confidence Interval of individual regression coefficient $\hat{\beta}_j$
- Pivot is

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{MS_{Res}C_{jj}}} \sim t_{n-p-1},$$

where C_{jj} is the diagonal element of $(X^T X)^{-1}$ matrix.

- A $100(1 - \alpha)\%$ CI for β_j is

$$\left[\hat{\beta}_j \pm t_{n-p-1, \alpha/2} \sqrt{MS_{Res}C_{jj}} \right].$$

Confidence interval for mean response:

- Let $x_0 = \begin{pmatrix} 1 \\ x_{01} \\ \vdots \\ x_{0p} \end{pmatrix}$ be a value of the regressor vector
- The mean response at x_0 is $x_0^T \beta$
- Pivot is $\frac{\hat{y}_0 - x_0^T \beta}{\sqrt{MS_{Res} x_0^T (X^T X)^{-1} x_0}} \sim t_{n-p-1}$
- A $100(1 - \alpha)\%$ CI for $x_0^T \beta$ is

$$\left[\hat{y}_0 \pm t_{n-p-1, \alpha/2} \sqrt{MS_{Res} x_0^T (X^T X)^{-1} x_0} \right]$$

Prediction Interval for New Observation

- Consider a level of regressor x_0 .
- Let y_0 be the corresponding value of the response.
- We want a prediction interval for y_0 .
- Let $\Psi = y_0 - \hat{y}_0 = y_0 - x_0^T \hat{\beta} \sim N(0, \sigma^2(1 + x_0^T (X'X)^{-1} x_0))$.
- A pivot is $\frac{\Psi - 0}{\sqrt{MS_{Res}(1 + x_0^T (X'X)^{-1} x_0)}} \sim t_{n-p-1}$.
- A $100(1 - \alpha)\%$ prediction interval of y_0 is

$$\left[\hat{y}_0 \mp t_{n-p-1; \frac{\alpha}{2}} \sqrt{MS_{Res}(1 + x_0^T (X'X)^{-1} x_0)} \right]$$

Model Adequacy Checking (Regression Diagnostics)

- Major assumptions
 - linear relationship
 - Error mean zero
 - Error variance is constant
 - Error are uncorrelated
 - Error are normally distributed
- Gross violation of the assumptions may lead to a totally different model with opposite conclusions.
- Unusual observations: Sometimes just a few observations do not fit the model. These few observations might change the choice and fit of the model.
- We perform the checking using residuals.

Different Residuals

- We now define 3 types of residuals.
- Residual: $e_i = y_i - \hat{y}_i$.
- Standardized residual: $d_i = \frac{e_i}{\sqrt{MS_{Res}}}$
- Studentized residual:
 - $e = (I - H)\epsilon \Rightarrow Var(e) = \sigma^2(I - H)$.
 - $Var(e_i) = \sigma^2(1 - h_{ii})$ and $Cov(e_i, e_j) = -\sigma^2 h_{ij}$.
 - It can be shown that $0 \leq h_{ii} \leq 1$.
 - $r_i = \frac{e_i}{\sqrt{MS_{Res}(1 - h_{ii})}}$.

Residual Plots: Q-Q Plot (Test for normality)

The residuals can be assessed for normality using a Q-Q plot. This compares the residuals to “ideal” normal observations.

We plot the quantiles corresponding to sorted residuals (e_i) against $\Phi^{-1}(\frac{i}{n+1})$ for $i = 1, \dots, n$.

Residual Plots: Q-Q Plot (Test for normality)

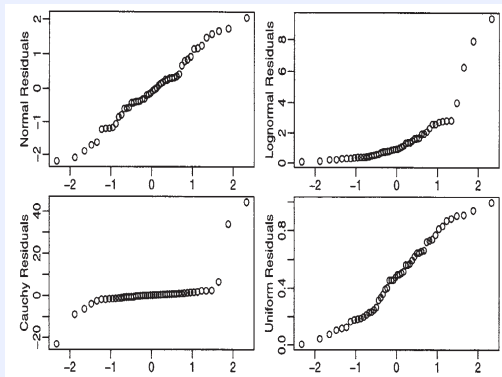
Four Q-Q plots are:

Normal: ideal

Lognormal: an example of a skewed distribution

Cauchy: an example of a long-tailed (platykurtic) distribution

Uniform: an example of a short-tailed (leptokurtic) distribution

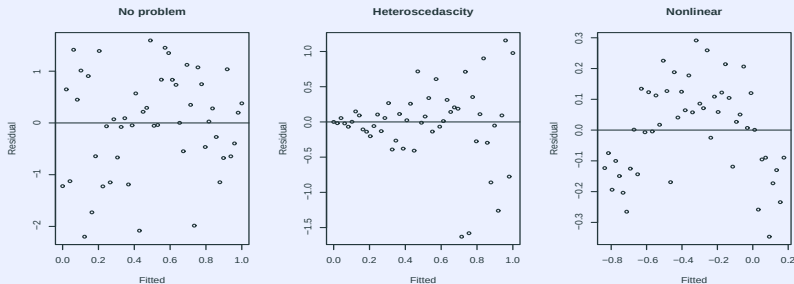


If non-Normal?

- For short-tailed distributions, the consequences of nonnormality are not serious and can reasonably be ignored.
- For skewed errors, a transformation of the response may solve the problem.
- For heavy-tailed errors, we might just accept the nonnormality and base the inference on the assumption of another distribution or use resampling methods such as the bootstrap.

Residual Plots: Plot of residual against Fitted values

Plot $\hat{y}_i$ vs e_i (or d_i or r_i): (Test of constant variance and non-linear relation)



Residuals vs Fitted plots - the **first** suggests **no change** to the current model while the **second** shows **non-constant variance** and the **third** indicates **some nonlinearity** which should prompt some change in the structural form of the model

Residual Plots

- Plot of residual against regressor
 - Plot x_{ij} vs $e_i \forall j$
 - In previous plot, replace 'fitted' by x_{ij} to find similar interpretation.
 - Repeat it for all the regressors.

If nonconstant variance?

- Two approaches
 - Weighted least squares
 - Transformation of variables
- Let h be a transformation that is applied on response

$$h(y) = h(Ey) + (y - Ey)h'(Ey) + \dots$$

$$\text{var } h(y) = h'(Ey)^2 \text{var } y + \dots$$

We ignore the higher order terms. For $\text{var } h(y)$ to be constant we need

$$h'(Ey) \propto (\text{var } y)^{-1/2},$$

which suggests

$$h(y) = \int \frac{dy}{\sqrt{\text{var } y}} = \int \frac{dy}{\text{SD}y}.$$

- If $\text{var } y = \text{var } \varepsilon \propto (Ey)^2$, then $h(y) = \log y$ is suggested.
- If $\text{var } \varepsilon \propto (Ey)$, then $h(y) = \sqrt{y}$.

Checking Correlated Errors

- Index plot: Plot residual against index (time).
- Plot e_i against e_{i-1} .
- Remedy: Use generalized least squares method

Checking the Structure of the Model

How do we check whether $E(y) = X\beta$ is correct?

Partial Regression Plot:

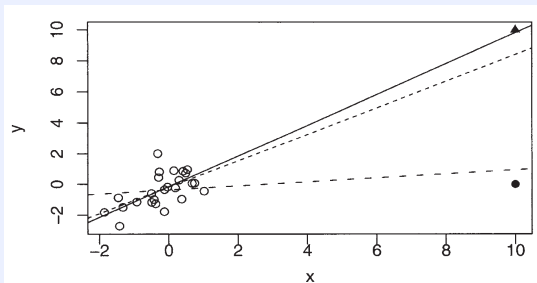
- Complete/correct marginal effect
- Let we want to study the marginal effect of x_j on y
- y is regressed on $x_1, \dots, x_k$ except x_j .
- x_j is regressed on $x_1, \dots, x_k$ except x_j .
- Plot y residuals against x_j residuals
- For ideal scenario, the partial regression plot should show a linear relationship (straight line with non-zero slope)

Remedy: Change you model and/or add other regressors.

Finding Unusual Observations

- Outliers: Observations that do not fit the model well.
- Influential observations: Observations that change the fit of the model in a substantive manner.
- A point can be none, one or both of these.
- Leverage points: Observations that are unusual in the predictor space. It also has the potential to influence the fit.

Unusual Observations



- The solid line is the fit including the ▲ point but not the ● point.
- The dotted line is the fit without both ▲ and ● points.
- The dashed line is the fit with the ● point but not the ▲ point.

Finding Outliers

- Order scaled residuals :

$$\frac{|e_i|}{\sqrt{\sum_{i=1}^n e_i^2}}$$

- The maximum one considerable larger (3 or 4 times) than others indicated potential outlier in y -space

Some points about outliers

- Two or more outliers next to each other can hide each other.
- An outlier in one model may not be an outlier in another when the variables have been changed or transformed.
- The error distribution may not be normal and so larger residuals may be expected.
- Individual outliers are usually much less of a problem in larger datasets.

- Check for a data-entry error first. These are relatively common. Unfortunately, the original source of the data may have been lost.
- Examine the physical context—why did it happen?
- Exclude the point from the analysis but try reincluding it later if the model is changed.
- Suppose you find outliers that cannot reasonably be identified as mistakes or aberrations, but are viewed as naturally occurring – Use robust estimation
- It is dangerous to exclude outliers in an automatic manner.

Leverage

- h_{ii} : the i -th diagonal of hat matrix H
- h_{ii} 's are called leverages
- Note that

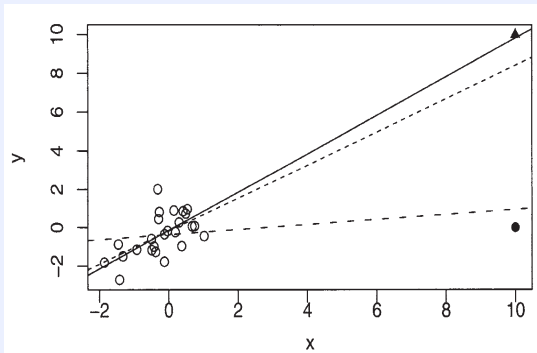
$$h_{ii} = x_i' (X'X)^{-1} x_i = \frac{1}{n} + \frac{MD_i^2}{n-1}$$

- $MD_i^2 = (x_i - \bar{x})^T S^{-1} (x_i - \bar{x})$: Mahalanobis distance
- Large values of h_{ii} indicates that the i -th observation is a leverage point

Leverage (continued)

- $\sum_{i=1}^n h_{ii} = p$ (as $\text{trace}(H) = \text{rank}(H) = \text{rank}(X) = p$)
- Average size of diagonals is $\frac{p}{n}$
- $h_{ii} > \frac{2p}{n}$: i -th observation is assumed to be leverage point
- Not all leverage points are going to be influential

Finding influential observations



Finding influential observations

- Cook's D :

$$\begin{aligned} D_i &= \frac{(\hat{\beta} - \hat{\beta}_{(-i)})'(X'X)(\hat{\beta} - \hat{\beta}_{(-i)})}{p \cdot \text{MSE}} = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(-i)})^2}{p \cdot \text{MSE}} \\ &= \frac{r_i^2}{p} \times \frac{h_{ii}}{1 - h_{ii}} \end{aligned}$$

- $\hat{\beta}_{(-i)}$: LSE of β obtained by deleting the i -th observation
- $\hat{y}_{j(-i)}$: Fitted value of y_i obtained by deleting the i -th observation
- r_i : i -th Studentized residual
- Rule of thumb: $D_i > 1$ indicates i -th point is influential point

The PRESS Statistic

- How do you measure prediction capability of a fitted model?
- Delete i^{th} observation. Fit the model on remaining $(n - 1)$ observations. Now, predict y_i . Let the corresponding prediction error be $e_{(i)} = y - \hat{y}_{(i)}$ (PRESS Residual).
- $e_{(i)} = \frac{e_i}{1-h_{ii}}$. (Prove it by yourself)
- $Var(e_{(i)}) = \frac{\sigma^2}{1-h_{ii}}$.
- Standardized PRESS residual is $\frac{e_{(i)}}{\sqrt{Var(e_{(i)})}} = \frac{e_i}{\sqrt{\sigma^2(1-h_{ii})}}$, which is standardized residual if we estimate σ^2 by MS_{Res} .

The PRESS Statistic (continued)

- $PRESS = \sum_{i=1}^n e_{(i)}^2 = \sum_{i=1}^n \left(\frac{e_i}{1-h_{ii}} \right)^2$.
- PRESS is a measure of how well a regression model perform in predicting new observations.
- R^2 for prediction :
 $R_{prediction}^2 = 1 - \frac{PRESS}{SST}$: gives indication of the prediction capability of the regression model.
- Using PRESS, we may compare model.

- In MLR, adding a variable to a model can increase the value of R^2 .
- To overcome this problem:

$$\text{Adjusted } R^2 : \quad R_{adj}^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2 / (n - p - 1)}{\sum_i (y_i - \bar{y})^2 / (n - 1)}.$$

Multicollinearity

- Near linear relationship among regressors ($\sum_{j=1}^p t_j x_j \simeq 0$)
- Effect of Multicollinearity:
 - Consider scaled response and regressor (length unit).
 - Consider $y = \beta_1 x_1 + \beta_2 x_2 + \epsilon$.
 - $\hat{\beta}_1 = \frac{r_{1y} - r_{12} r_{2y}}{1 - r_{12}^2}$, and $\hat{\beta}_2 = \frac{r_{2y} - r_{12} r_{1y}}{1 - r_{12}^2}$.
 - $Var(\hat{\beta}_j) = \frac{\sigma^2}{1 - r_{12}^2}$, $Cov(\hat{\beta}_1, \hat{\beta}_2) = \frac{-r_{12} \sigma^2}{1 - r_{12}^2}$.
 - If $|r_{12}| \rightarrow 1$, $Var(\hat{\beta}_j) \rightarrow \infty$, and $|Cov(\hat{\beta}_1, \hat{\beta}_2)| \rightarrow \infty$.

- Effect of Multicollinearity (contd.):

- $L_1^2 = (\hat{\beta} - \beta)^T (\hat{\beta} - \beta).$
- $E(L_1^2) = \sum_{j=1}^p Var(\hat{\beta}_j) = \sigma^2 Tr(X'X)^{-1} = \sigma^2 \sum_{j=1}^p \frac{1}{\lambda_j},$ where λ_j 's are eigenvalues of $(X'X).$
- If $(X'X)$ is ill-conditioned then at least one λ_j will be small $\Rightarrow E(L_1^2)$ is big.
- Therefore, we have $E(\hat{\beta}^T \hat{\beta}) = \beta' \beta + \sigma^2 Tr(X'X)^{-1},$ implies magnitude of $\hat{\beta}$ are large.

Multicollinearity Diagnostics

- Examination of correlation matrix ($X'X$):
 - If x_i and x_j are nearly linearly dependent, then $|r_{ij}|$ should be close to 1.
 - However, this procedure is helpful to detect near linear dependence between a pair of regressors only.

Multicollinearity Diagnostics (Contd.)

- Variance Inflation Factors (VIFs):

- It can be shown that

$$Var(\hat{\beta}_j) = \sigma^2 C_{jj} = \frac{\sigma^2}{1 - R_j^2},$$

where R_j^2 is the coefficient of determination obtained when x_j is regressed on remaining $(k - 1)$ regressors.

- $VIF_j = C_{jj}$: This measures the factor by which the variance of $\hat{\beta}_j$ inflated due to the near linear dependence.
 - Rule of thumb : If any of $VIF > 5$, the associated coefficient is estimated poorly due to multicollinearity.

Multicollinearity Diagnostics (Contd.)

- Eigen System Analysis of $(X'X)$:

- The eigen values, $\lambda_1, \lambda_2, \dots, \lambda_p$, can be used to see the extent of multicollinearity.
- Small eigen values (one or more) $\Rightarrow$ multicollinearity.
- Condition number $= \kappa = \frac{\lambda_{max}}{\lambda_{min}}$.
- Rule of thumb :

$\kappa < 100 \rightarrow$ No serious problem with multicollinearity.

$100 \leq \kappa < 1000 \rightarrow$ moderate to strong multicollinearity.

$\kappa \geq 1000 \rightarrow$ severe multicollinearity.

- Condition indices : $\kappa_j = \frac{\lambda_{max}}{\lambda_j}$, $j = 1, 2, \dots, p$

$\kappa_j \geq 1000 \rightarrow$ provide useful information on the number of near linear dependence.

Singular Value Decomposition (SVD)

Let $\mathbf{X}$ be the $n \times p$ matrix of regressors (usually centered and scaled).
The SVD of $\mathbf{X}$ is:

$$\mathbf{X} = \mathbf{U}\mathbf{\Gamma}\mathbf{V}^T$$

where:

- $\mathbf{U}$ is an $n \times p$ matrix such that $\mathbf{U}^T\mathbf{U} = \mathbf{I}_p$.
- $\mathbf{V}$ is a $p \times p$ orthogonal matrix ($\mathbf{V}^T\mathbf{V} = \mathbf{I}_p$). The columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{X}^T\mathbf{X}$.
- $\mathbf{\Gamma}$ is a $p \times p$ diagonal matrix of **singular values** $\mu_1, \mu_2, \dots, \mu_p$.

Note: The eigenvalues of $\mathbf{X}^T\mathbf{X}$ are $\lambda_j = \mu_j^2$.

Condition Number and Condition Indices

The **Condition Number** of the matrix $\mathbf{X}$ is defined as:

$$\eta = \frac{\mu_{\max}}{\mu_{\min}}$$

The **Condition Indices** of the matrix $\mathbf{X}$ as:

$$\eta_j = \frac{\mu_{\max}}{\mu_j}, \quad j = 1, 2, \dots, p$$

Thresholds for η_j

- $\eta_j < 30$: Weak dependency.
- $30 < \eta_j < 100$: Moderate to strong dependency.
- $\eta_j > 100$: Severe multicollinearity.

Variance Decomposition Proportions

The covariance matrix of $\hat{\beta}$ is $\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$. Using SVD:

$$\text{Var}(\hat{\beta}_j) = \sigma^2 \sum_{k=1}^p \frac{v_{jk}^2}{\mu_k^2}$$

The **Variance Decomposition Proportion** (VDP) for the j -th regression coefficient associated with the k -th component is:

$$\pi_{kj} = \frac{v_{jk}^2 / \mu_k^2}{\sum_{i=1}^p v_{ji}^2 / \mu_i^2}$$

- π_{kj} represents the proportion of the variance of $\hat{\beta}_j$ explained by the k -th singular value.

Diagnostic Rule

A multicollinearity problem is diagnosed when:

1. A condition index η_k is large (typically > 30).
2. Two or more regressors have large variance decomposition proportions π_{kj} (typically > 0.5) associated with that large η_k .

Method for dealing with multicollinearity

- Source of multicollinearity:
 - Data collection method → collecting more data.
 - Constraints in model or population → Model respecification
 - Model specification → Model respecification
 - An overdefined model → Model respecification, and other method of estimate (like Ridge regression).